StructFuse-Net: A Structure-Aware Multimodal Fusion Network for Geometry-Consistent Optical–SAR Image Segmentation

Xin Li, Xianbin Wen, Haixia Xu, and Xinyu Wang

***Abstract*—Multimodal remote sensing segmentation remains challenging due to modality discrepancies, structural inconsistencies, and sensor-induced noise, which often lead to blurred boundaries and geometric distortions. To address these issues, we propose StructFuse-Net, a structure-aware multimodal fusion framework for geometry-consistent optical–synthetic aperture radar (SAR) image segmentation. The framework comprises three complementary modules: 1) a structure-driven directional fusion (SDDF) module that enhances low-level representations by suppressing speckle noise and reinforcing geometric cues through cross-modal attention and wavelet-based directional decomposition; 2) a semantic refinement and alignment fusion (SRAF) module that improves high-level consistency by refining channel–spatial responses and aligning cross-modal semantics via residual guidance and frequency-aware enhancement; and 3) a structure-aware cross-level fusion (SACLF) module that connects low- and high-level features through adaptive gating and orthogonal directional encoding, ensuring multiscale geometry-consistent feature representation. Extensive experiments on the WHU-OPT-SAR and YESeg-OPT-SAR datasets demonstrate that StructFuse-Net consistently outperforms state-of-the-art (SOTA) baselines, achieving overall accuracies of 82.7% and 81.5% with mean intersection over union (mIoU) of 54.5% and 62.4%, respectively. Moreover, experiments on a cloud-disturbed subset of the Pohang dataset further validate the robustness of StructFuse-Net under severe optical degradation, where it maintains coherent geometry and boundary continuity with smaller performance degradation than competing methods. Beyond quantitative improvements, the proposed network delivers superior boundary fidelity and spatial coherence, particularly for elongated and structure-sensitive objects such as roads and rivers. These results indicate that StructFuse-Net provides a robust and generalizable solution for geometry-consistent optical–SAR semantic segmentation.**

***Index Terms*—Boundary refinement, cross-level feature fusion, image segmentation, multimodal remote sensing, optical–synthetic aperture radar (SAR) fusion, structure-aware fusion, wavelet decomposition.**

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I. Introduction

REMOTE sensing image (RSI) segmentation is a fundamental task in Earth observation, supporting a wide range of applications such as urban planning [1], environmental monitoring [2], and disaster assessment [3]. Despite remarkable progress, most existing approaches still rely on single-modal imagery—such as optical, synthetic aperture radar (SAR), multispectral, or hyperspectral data—whose limited information diversity often causes performance degradation in complex real-world environments [4]. In contrast, multimodal semantic segmentation leverages complementary cues across heterogeneous sources to produce more robust and comprehensive land-cover maps [5]. Among diverse modality combinations, optical–SAR fusion has become one of the most extensively explored settings [6]. Optical imagery offers fine spatial resolution and rich texture but is sensitive to clouds, shadows, and illumination variation [7]. Conversely, SAR imagery provides all-weather, day-and-night imaging with superior geometric and structural sensitivity, but suffers from speckle noise and limited texture representation [8]. These inherent sensing characteristics give rise to modality discrepancies, structural inconsistencies, and sensor-induced noise, which collectively result in blurred boundaries and geometric distortions in segmentation maps. Modality discrepancies primarily originate from the fundamentally different imaging mechanisms of optical and SAR sensors. As illustrated in Fig. 1(a), optical images capture rich spectral and textural details but are vulnerable to cloud occlusion and specular reflections, which obscure land-cover boundaries. In contrast, SAR images are affected by geometric distortions such as foreshortening, where sloped surfaces (e.g., mountainous terrain or building rooftops) appear compressed, leading to shape inconsistencies when directly fused with optical imagery. Structural inconsistencies further arise from the distinct ways in which the two modalities represent spatial structures. Optical imagery emphasizes fine edges and local textures, which can be lost in shadowed or low-contrast regions [Fig. 1(b)], whereas SAR imagery preserves global structural topology but tends to blur fine boundaries due to speckle interference. In addition, sensor-induced noise exacerbates these challenges. Optical images are influenced by atmospheric scattering that reduces interclass contrast, while SAR imagery suffers from multiplicative speckle noise, obscuring subtle structures such as river edges and producing jagged contours. Fig. 1(b) further

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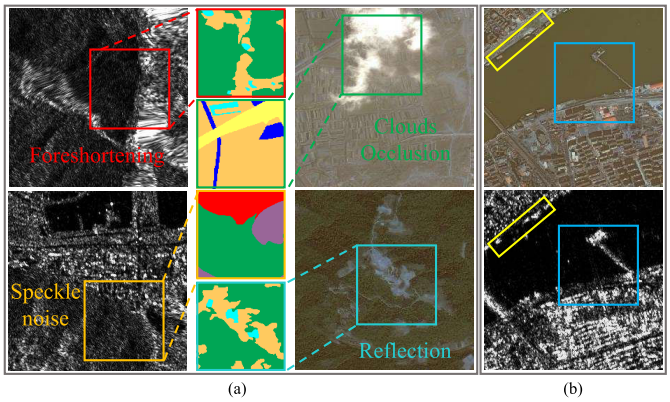


Fig. 1. Modality discrepancies and fusion challenges in optical–SAR RSI segmentation. (a) Illustration of modality-specific limitations. Optical imagery is affected by cloud occlusion (green boxes), reflection (blue boxes), and illumination variation, while SAR imagery suffers from speckle noise (orange boxes) and geometric distortions such as foreshortening (red boxes). (b) Pair of coregistered optical and SAR images from the same region, showing their complementary spectral, textural, and structural characteristics.

highlights the complementary nature of the two modalities, where optical data compensate for the limited textural expressiveness of SAR imagery and SAR data reinforce the structural integrity of optical imagery. This complementarity underpins the motivation for geometry-consistent multimodal fusion in optical–SAR remote sensing segmentation [9]. Notably, geoscience applications impose stringent requirements on the fusion results, which directly drive the technical design of our framework. For urban planning [10], the geometric topological consistency of roads and buildings is indispensable for assessing construction compliance and transportation network connectivity. In water resource monitoring [11], the boundary robustness of rivers, lakes, and wetlands against noise (e.g., SAR speckle and optical cloud shadows) ensures accurate estimation of water storage capacity and basin delineation. For disaster impact assessment (e.g., flood and earthquake) [12], the semantic accuracy of damaged areas under adverse imaging conditions (e.g., heavy rainfall and smoke) is critical for rapid rescue and resource allocation. These geoscience-specific demands necessitate that multimodal fusion not only addresses technical challenges but also aligns with practical application scenarios.

In particular, geometric discrepancies between optical and SAR imagery can lead to shape distortions and boundary misalignment, which are compounded by sensor-specific noise and structural inconsistencies [13]. These challenges make geometry-consistent fusion critical for preserving spatial topology and structural integrity across modalities. Additionally, semantic drift across modalities further complicates the segmentation task, particularly for objects with complex boundaries or small, elongated features. These issues are of particular importance in geoscience applications, where accurate land-cover classification relies on preserving the geometry of natural and man-made structures. Multimodal fusion in this context not only needs to address the technical challenges but also needs to directly benefit practical applications, such as detailed mapping of urban environments, monitoring of vegetation cover, and assessment of disaster impacts.

Recent research has investigated multimodal fusion from multiple perspectives. Low-level fusion methods, such as SFCFusion [14] and SOSTF [15], enhance structural and textural details through spatial–frequency decomposition and structure–texture separation. Although effective for local detail preservation, these methods remain vulnerable to misalignment and modality-specific noise, especially SAR speckle. High-level fusion approaches, including MICF-Net [16] and MPLSeg [17], employ cross-modal attention or frequency-domain modulation to enhance semantic consistency, yet excessive abstraction often leads to boundary ambiguity and inconsistent category activation. Cross-level fusion frameworks, such as PACSCNet [18], attempt to reconcile semantic abstraction with spatial precision via hierarchical refinement, but the absence of explicit structural priors still results in blurred boundaries and suboptimal delineation of elongated or fine-grained objects.

To overcome these limitations, we propose StructFuse-Net, a hierarchical cross-modal fusion framework that progressively integrates optical and SAR features from shallow to deep layers. StructFuse-Net is designed to address specific challenges in multimodal fusion, including modality mismatch, noise suppression, geometric inconsistencies, boundary precision, and cross-level semantic and structural alignment. The structure-driven directional fusion (SDDF) module tackles modality mismatch at the low-level fusion stage by leveraging modality-specific structural priors and direction-aware frequency decomposition, which enhances geometric cues (edges, gradients, and textures) while suppressing noise and mitigating semantic interference, ensuring better alignment of structural information between optical and SAR modalities. The SDDF module also effectively addresses noise suppression by reducing SAR speckle noise and preserving fine-grained details from both modalities. For geometric inconsistencies, the SDDF module applies direction-sensitive decomposition to ensure proper alignment of structural cues, particularly for objects with anisotropic geometries, such as roads and rivers. Moving to the high-level fusion stage, the semantic refinement and alignment fusion (SRAF) module adopts an adaptive "refine-then-fuse" strategy that refines and aligns semantic responses through dual-dimensional attention and frequency-aware enhancement, ensuring category-level coherence and sharper boundaries, especially for fine-grained or elongated targets. Finally, the structure-aware cross-level fusion (SACLF) module bridges shallow and deep representations by establishing bidirectional structural–semantic interaction via adaptive gating and direction-sensitive convolution, thereby maintaining cross-scale spatial-semantic consistency and recovering fine-grained boundaries while retaining high semantic accuracy.

Extensive experiments on benchmark optical–SAR datasets demonstrate that StructFuse-Net significantly improves segmentation accuracy, particularly for thin, elongated, and boundary-sensitive objects. The results confirm that integrating structure-driven low-level enhancement, high-level semantic alignment, and cross-level structural guidance within a unified framework effectively achieves geometry-consistent multimodal segmentation.

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The main contributions of this work can be summarized as follows.

1) A structure-aware multimodal fusion framework for geometry-consistent optical–SAR segmentation. We propose StructFuse-Net, which explicitly embeds geometric and structural priors into multimodal feature fusion, enabling geometry-consistent alignment across optical and SAR modalities and effectively mitigating spatial distortion and semantic drift observed in conventional fusion strategies.
2) Hierarchical modality adaptive feature enhancement for joint optimization of geometric structure and semantic representation. We design complementary low- and high-level fusion mechanisms, where the SDDF module preserves structural priors and suppresses SAR speckle through bidirectional cross-modal attention and directional frequency decomposition, and the SRAF module refines high-level semantics and sharpens object boundaries via a refine-then-fuse strategy.
3) Cross-scale structure semantic interaction with adaptive geometry-aware optimization. We introduce the SACLF module to establish bidirectional guidance between low- and high-level features through adaptive gating and directional convolution and further incorporate a multiobjective adaptive loss to enforce geometry-consistent learning across modalities and scales.

The remainder of this article is organized as follows. Section II reviews recent advances in multimodal remote sensing segmentation, with a focus on noise-resilient fusion, semantic consistency alignment, and structural consistency modeling. Section III presents the proposed StructFuse-Net in detail, describing its three core modules, as well as the overall loss formulation. Section IV provides the experimental setup, evaluation metrics, and comprehensive results, including ablation studies, visual comparisons, and complexity analysis. Section V concludes this article and discusses future directions toward geometry-consistent multimodal fusion and physically informed remote sensing models.

II. Related Work

A. Noise-Resilient Fusion

Low-level features such as edges, gradients, and textures are crucial for delineating fine-grained land-cover structures. However, optical and SAR data exhibit distinct noise characteristics. Optical imagery is easily affected by clouds, shadows, and illumination variations [19], whereas SAR data are corrupted by speckle arising from coherent scattering [20]. These modality-specific disturbances degrade feature fidelity and hinder cross-modal fusion.

Early clustering-based fusion methods integrated low-level cues while implicitly suppressing isolated noise [21], yet the lack of explicit noise modeling led to performance degradation under severe corruption. Ye et al. [22] introduced visual-saliency guidance to highlight informative regions, but the reliability of saliency estimation itself became a bottleneck, and cross-modal structural consistency remained weak. Chen et al. [23] proposed a separation-and-aggregation gate to filter and recalibrate modality-specific features before fusion, reducing noise but failing to preserve directional geometric cues for segmenting thin, linear features. Hosseinpour et al. [24] adopted a gated fusion module (GFM) and top-down semantic refinement for multilevel cue integration, enhancing robustness in building extraction, but lacked explicit structural constraints, resulting in blurred boundaries in fine-grained categories like village edges and sparse vegetation.

Subsequent research explored frequency-domain solutions. Wavelet-based SFFNet [25] disentangled structural information from noise but neglected directional cues within sub-bands, leaving geometry-aware enhancement incomplete. Attention-based designs such as AFF [26] and CBAM-based models reweighted shallow features to emphasize shared structures; however, the absence of frequency or orientation priors limited geometric fidelity under strong speckle. Quan et al. [27] attempted implicit suppression by adaptively upweighting salient geometries, yet the lack of dedicated speckle modeling caused performance collapse when clutter dominated.

Collectively, these studies demonstrate that robust, modality-aware low-level enhancement is not a postprocessing luxury but an essential prerequisite for reliable multimodal segmentation, providing the foundation on which subsequent high-level fusion must build.

B. Semantic Consistency Fusion

Maintaining semantic consistency across modalities remains a fundamental challenge in multimodal remote sensing segmentation. Optical features emphasize spectral and textural content, while SAR features encode scattering and structural properties, often leading to class activation drift and boundary misalignment [28].

Early alignment-constrained approaches [29] enforced category-level consistency to mitigate semantic drift, but their coarse alignment overlooked fine-grained semantics. CLAN [30] extended this concept via category-adversarial reweighting, yet its dual-classifier mechanism was unstable when domain divergence increased. Disentangled representation learning [31] further separated content and style features, but its stochastic style code struggled to represent small objects and complex boundaries, producing unstable results. Semantic dependency modeling, such as DPGUNet [32], improved cross-scale coherence without alignment labels but relied on a fixed superpixel graph, limiting adaptability to varying granularities.

Recent methods have shifted toward progressive or hierarchical alignment. Progressive semantic alignment [28] gradually calibrates optical features toward SAR representations at both domain and category levels, while DecAlign [33] decouples modality-specific and shared semantics prior to fusion. Nevertheless, cosine-similarity alignment still permits information leakage across latent spaces. Ma et al. [34] employed cross-modal transformers and multiscale augmentation to capture long-range dependencies but overrelied on global modeling, which affected local geometric fidelity. StructFuse-Net mitigates this with dual-branch fusion. Ma et al. [35] combined a visual state-space (VSS) auxiliary branch with a convolutional main branch to model long-range

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dependencies but lacked directional geometric encoding, which led to inconsistent segmentation under strong speckle. GMFANet [36] introduced gradient-guided spatial attention to enhance salient targets, but it primarily refines saliency rather than enforcing cross-modal semantic coherence. FTransUNet [37] adopted a hybrid architecture of CNNs and transformers to enhance feature representation with global contextual modeling, but it struggled with boundary ambiguity in category-transition regions.

Overall, these studies reveal that ensuring hierarchical semantic alignment between modalities is critical for accurate and robust multimodal land-cover segmentation, rather than serving as an optional refinement.

C. Structural Consistency Fusion

Preserving geometric fidelity and boundary continuity is vital for segmenting thin or elongated objects such as roads, rivers, and building edges. However, integrating complementary structural cues from optical and SAR imagery remains difficult due to disparities in spatial resolution, semantic abstraction, and modality-dependent noise [38].

Early semantic-guided fusion frameworks, such as the multilevel cue integration of Li et al. [39] and edge refinement by Cui et al. [40], improved local structural clarity but did not explicitly enforce geometric consistency across modalities. Deeplabv3 + [41] used atrous separable convolution and a decoder for boundary refinement in single-modal segmentation but lacked cross-modal fusion, which limited its ability to handle modality-specific discrepancies like optical blurring and SAR speckle. Shen et al. [42] adopted a multiscale pyramid to capture both global and local contexts, yet the fused representations still suffered from blurred boundaries in the absence of explicit structural constraints. Attention-based approaches, including the multimodal transformer proposed by Bahaduri et al. [43], enhanced structure-aware features but failed to guarantee spatial alignment between modalities.

Recent studies on multimodal optical–SAR change detection indicate that explicitly modeling structural consistency across modalities can alleviate cross-modal misalignment, especially in the presence of large appearance discrepancies. Representative approaches have explored this issue from different perspectives. Graph-based modeling strategies exploit nonlocal patch similarity to enforce structural correspondence across modalities [44], while geometry-aware formulations introduce explicit geometric constraints to regulate cross-modal deformation [45]. In addition, locality-preserving structural modeling has been shown to maintain spatial coherence between optical and SAR representations [46]. Although change detection and semantic segmentation address different tasks, these studies collectively suggest that structural priors play a critical role in regularizing cross-modal representations under heterogeneous sensing conditions. Building on these insights, the SACLF module in our framework incorporates directional structural encoding to support geometry-consistent feature fusion.

Beyond spatial-domain modeling, several studies have explored frequency-domain representations for multimodal fusion. Zhang et al. [47] employed the discrete wavelet transform (DWT) to decompose multimodal features into directional sub-bands, which helps enhance edge and texture information while preserving coarse semantic structure. Together, these studies suggest that boundary preservation in multimodal segmentation benefits from a combination of multiscale processing, attention mechanisms, and structural constraints. However, how to effectively integrate these factors to achieve geometry-consistent representations across heterogeneous modalities remains an open problem, motivating the cross-level, geometry-aware fusion strategy adopted in StructFuse-Net.

III. Proposed Method

A. Overall Structure of the StructFuse-Net

To address modality discrepancy and structural inconsistency in multimodal RSI segmentation, we propose StructFuse-Net, a structure-aware fusion framework for robust and geometry-consistent optical–SAR segmentation. Unlike conventional fusion strategies that process multimodal features in a uniform manner, StructFuse-Net is designed under a hierarchical geometric-consistency principle, where modality-specific distortions are progressively corrected at their corresponding feature levels. As illustrated in Fig. 2, the overall architecture adopts a pyramid dual-stream encoder–decoder structure to hierarchically integrate complementary information from optical and SAR modalities. The complete workflow is summarized in Algorithm 1 to clarify the forward inference and training procedures.

Specifically, the encoder is composed of two parallel branches that independently extract hierarchical representations from optical and SAR inputs. The optical branch focuses on fine spatial textures and spectral reflectance patterns but is susceptible to illumination variations and boundary degradation. In contrast, the SAR branch captures anisotropic geometric structures under diverse backscattering conditions, while being affected by speckle-induced fragmentation. Owing to these heterogeneous sensing mechanisms, the two modalities exhibit misaligned structural cues, inconsistent boundary representations, and semantic drift across feature hierarchies. Accordingly, StructFuse-Net integrates optical and SAR features in a progressive manner through three geometrically motivated fusion modules, each designed to address the dominant inconsistency at a specific feature level.

At the low-level fusion stage, the SDDF module performs structure-driven correction by jointly exploiting bidirectional cross-modal attention and direction-aware frequency decomposition. By leveraging the complementary strengths of optical texture richness and SAR structural stability, SDDF recovers missing edges, suppresses speckle noise, and preserves directional cues of linear structures such as roads and rivers.

At the high-level fusion stage, the SRAF module conducts semantic refinement and alignment through dual-dimensional attention, residual semantic anchoring, and wavelet-based high-frequency enhancement. This stage mitigates semantic drift and boundary blur caused by heterogeneous high-level abstractions, leading to semantically coherent and boundary-precise optical–SAR representations.

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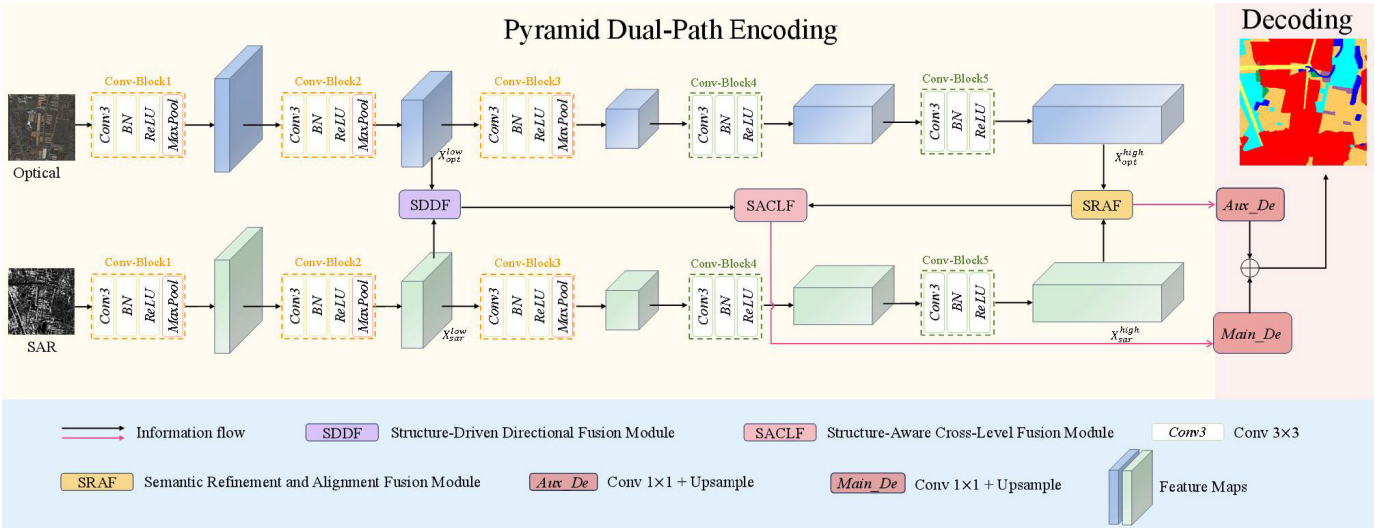


Fig. 2. Overall architecture of the proposed StructFuse-Net, a structure-aware multimodal fusion network for optical–SAR RSI segmentation. The model employs a pyramid dual-path encoder to extract hierarchical features from optical and SAR modalities, followed by three dedicated fusion modules: 1) the SDDF module for low-level structure-preserving enhancement; 2) the SRAF module for high-level semantic alignment; and 3) the SACLF module for cross-scale structural continuity and geometry-consistent integration. The decoder jointly reconstructs fine boundaries and semantic regions through multibranch deep supervision.

Algorithm 1 Workflow of the Proposed StructFuse-Net

Input:

Input Optical image: $X_{opt} \in \mathbb{R}^{4\times H\times W}$

Input SAR image: $Y_{sar} \in \mathbb{R}^{1\times H\times W}$

Label: $L \in \mathbb{R}^{H\times W}$

T_{train}: training phase

Output: The semantic segmentation results: $O_{final} \in \mathbb{R}^{H\times W}$

while T_{train} **do**

  Extract Features:

    OPT branch: Extract hierarchical features $F_{\text{opt}}^{\text{low}}$ and $F_{\text{opt}}^{\text{high}}$ using a dual-stream encode.

    SAR branch: Extract hierarchical features $F_{\text{sar}}^{\text{low}}$ and $F_{\text{sar}}^{\text{high}}$ using a dual-stream encode.

  Feature fusion:

    First, apply the **SDDF** module to fuse low-level features $X_{\text{opt}}^{\text{low}}, X_{\text{sar}}^{\text{low}}$, yielding $F_{\text{fused}}^{\text{low}}$.

    Then, apply the **SRAF** module to fuse X_{opt}^{high} and $X_{\text{sar}}^{\text{high}}$ through refinement and frequency-domain enhancement, producing $F_{\text{fused}}^{\text{high}}$.

    Finally, apply the **SACLF** module to integrate $F_{\text{fused}}^{\text{high}}$ and $F_{\text{fused}}^{\text{low}}$ via bidirectional gating and orthogonal directional encoding, producing F_{fused}.

  Optimization:

    Calculate the Loss function of StructFuse-Net: (Eq. 25)

    Calculate the segmentation loss: L_{ce}

    Calculate the Dice loss: L_{Dice}

    Calculate the Focal loss: L_{Focal}

    Calculate the boundary loss: L_{Boundary}

    Compute adaptive weights: $\alpha_i(k)$ (Eq. 30)

  return O_{final}

At the cross-level fusion stage, the SACLF module establishes bidirectional information propagation between structure-rich low-level features and semantic-rich high-level features. Through content-adaptive gating and orthogonal directional encoding, SACLF compensates for geometric information lost during semantic abstraction and preserves structural continuity across scales, thereby enhancing the representation of elongated and anisotropic objects.

Moreover, StructFuse-Net employs a hierarchical multibranch deep supervision strategy to jointly optimize the main and auxiliary outputs. This strategy stabilizes gradient propagation, encourages multiscale feature alignment, and improves segmentation robustness under diverse sensing conditions.

Overall, StructFuse-Net constitutes a complete geometry-preserving fusion pipeline. SDDF enforces low-level structural alignment, SRAF ensures high-level semantic coherence, and SACLF achieves cross-level geometric continuity. These stages are progressively coupled and jointly contribute to accurate and generalizable multimodal segmentation in complex remote sensing scenarios.

B. SDDF Module

Low-level features encode rich textural, boundary, and spatial details that are essential for accurately segmenting linear and fine-grained structures such as roads, rivers, and building contours. However, the heterogeneous imaging mechanisms of optical and SAR modalities result in distinct feature characteristics and pose fundamental challenges for low-level fusion. Specifically, optical imagery provides fine-grained textures and sharp edges but is highly sensitive to illumination variations and atmospheric effects, often leading to missing or distorted local boundary cues. In contrast, SAR imagery offers stable all-weather observations but suffers from speckle noise, which corrupts anisotropic backscattering signals that encode directional geometric information. As a result, shallow representations from the two modalities differ substantially: optical features emphasize spectral–textural richness, whereas SAR features encode directional geometric structures that are

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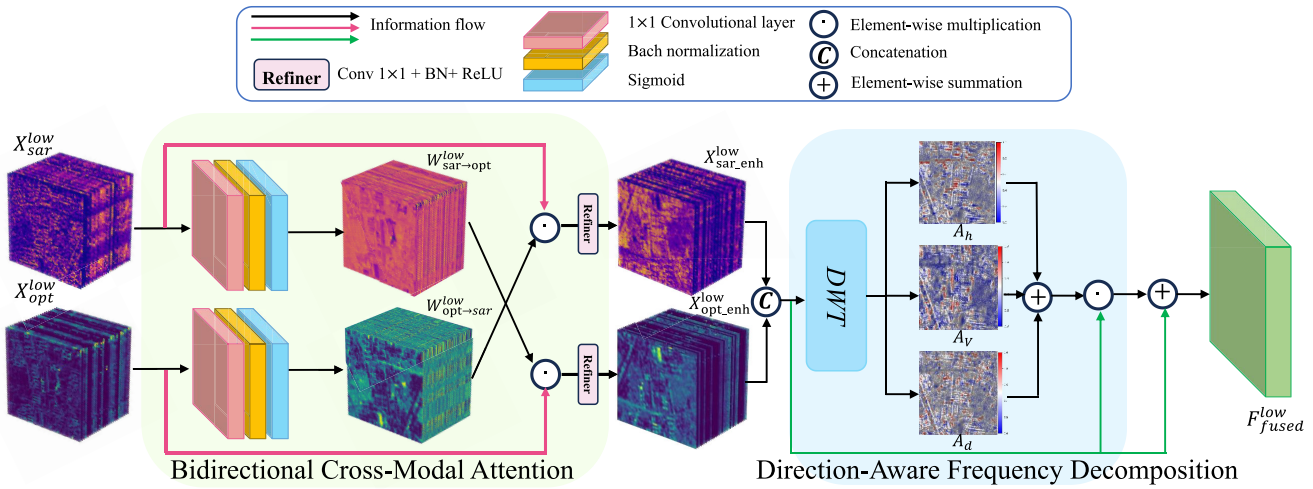


Fig. 3. SDDF module. The SDDF module consists of two main components: 1) a bidirectional cross-modal attention mechanism that enables optical and SAR features to mutually guide and correct each other, producing enhanced modality-specific features ($X^{\text{low}}_{\text{opt,enh}}$ and $X^{\text{low}}_{\text{sar,enh}}$) and 2) a direction-aware frequency enhancement stage, where DWT decomposes the fused representation into horizontal (A_h), vertical (A_v), and diagonal (A_d) sub-bands. These orientation-specific components are adaptively aggregated to form the final fused feature $F^{\text{low}}_{\text{fused}}$, ensuring geometrically consistent and structurally enriched low-level representations.

partially degraded by noise, hindering reliable cross-modal alignment at the low feature level.

To address these dual challenges—cross-modal geometric misalignment and modality-specific noise corruption—and to reconcile the complementary yet conflicting properties of optical and SAR low-level features, we design the SDDF module (Fig. 3). The SDDF module aims to preserve modality-specific structural priors, suppress noise interference, and maintain geometric consistency across modalities. To this end, SDDF enhances low-level fusion through two core mechanisms: bidirectional cross-modal attention and direction-aware frequency decomposition, both of which are explicitly tailored to the geometric characteristics of optical–SAR data and their anisotropic edge structures.

1) Bidirectional Cross-Modal Attention: Conventional unidirectional attention mechanisms typically perform one-way feature weighting and thus fail to achieve mutual error correction between modalities, making them ineffective for handling the asymmetric noise characteristics and feature deficiencies of optical and SAR data. To enable robust cross-modal interaction, the proposed SDDF module introduces a bidirectional cross-modal attention mechanism that explicitly models mutual correction between modalities at the low feature level.

Given the shallow features $X^{\text{low}}_{\text{opt}}$ and $X^{\text{low}}_{\text{sar}}$ extracted from optical and SAR encoders, two projection functions generate attention maps

$$W^{\text{low}}_{\text{sar}\to\text{opt}} = \sigma\left(F^{\text{low}}_{\text{sar}\to\text{opt}} X^{\text{low}}_{\text{sar}}\right) \tag{1}$$

$$W^{\text{low}}_{\text{opt}\to\text{sar}} = \sigma\left(F^{\text{low}}_{\text{opt}\to\text{sar}} X^{\text{low}}_{\text{opt}}\right) \tag{2}$$

where $\sigma(\cdot)$ denoted the sigmoid activation function. Unlike conventional attention that applies unidirectional modulation, the proposed bidirectional formulation enables reciprocal structural correction. SAR-driven attention weights reinforce blurred or missing edge cues in optical features caused by atmospheric interference, while optical-driven attention suppresses speckle-contaminated responses in SAR features by emphasizing texture-consistent structural regions. The attended features are then modulated element-wise as

$$X^{\text{low}}_{\text{sar_enh}} = X^{\text{low}}_{\text{sar}} \cdot W^{\text{low}}_{\text{opt}\to\text{sar}} \tag{3}$$

$$X^{\text{low}}_{\text{opt_enh}} = X^{\text{low}}_{\text{opt}} \cdot W^{\text{low}}_{\text{opt}\to\text{sar}}. \tag{4}$$

Through this bidirectional cross-modal interaction, salient edges and texture-consistent structures are selectively enhanced, while redundant or noisy responses are suppressed, establishing an initial layer of geometric alignment and noise reduction for subsequent fusion stages.

2) Direction-Aware Frequency Decomposition: After feature enhancement, lightweight 1×1 convolutions followed by batch normalization and ReLU activation are applied to reduce redundancy and unify feature statistics.

The two enhanced features are concatenated and fused into an intermediate representation F_{inter}. To better capture geometric structures aligned with physical orientations, SDDF decomposes F_{inter} via DWT into horizontal Y_h, vertical Y_v, and diagonal Y_d sub-bands, allowing direction-specific processing of edges and gradients

$$A_h = \sigma(f_h(Y_h)) \tag{5}$$

$$A_v = \sigma(f_v(Y_v)) \tag{6}$$

$$A_d = \sigma(f_d(Y_d)) \tag{7}$$

where $f_\theta(\cdot)$ denotes lightweight orientation-specific convolution and $\theta \in \{h, v, d\}$ represents horizontal, vertical, and diagonal directions, respectively. This decomposition is motivated by the anisotropic geometric characteristics of key land-cover targets, such as roads, rivers, and urban structures, which often exhibit locally smooth geometric patterns along dominant orientations. By explicitly separating directional components, DWT is well suited for isolating robust directional cues that remain stable under illumination variations in

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optical imagery. Meanwhile, SAR backscattering responses are inherently anisotropic due to imaging geometry, and wavelet decomposition naturally disentangles such directional structures. This allows the model to amplify geometry-consistent responses while suppressing speckle noise that is diffusely distributed across high-frequency components without consistent orientation. The resulting directional attention maps are aggregated to recalibrate the fused representation

$$F_{\text{fused}}^{\text{low}} = F_{\text{inter}} \cdot (1 + A_h + A_v + A_d). \tag{8}$$

Through direction-aware frequency decomposition and bidirectional cross-modal attention, SDDF reinforces linear continuity and boundary integrity in complex scenes, transforming calibrated features into a geometrically aligned low-level representation. Rather than acting as a simple feature enhancement block, SDDF establishes a geometry-aware alignment mechanism that explicitly models anisotropic scattering and directional structures across modalities. By jointly leveraging structural correction and frequency-domain decomposition, SDDF bridges the physical sensing gap between optical reflectance and SAR backscattering, providing a structurally consistent foundation for subsequent semantic fusion.

C. SRAF Module

High-level semantic fusion between optical and SAR features remains a major challenge in multimodal RSI segmentation, primarily due to semantic drift and boundary inconsistency. At this stage, the network transitions from low-level structural cues, such as edges and textures, to abstract category-level semantics, which requires accurate modeling of contextual dependencies and reliable cross-modal alignment. In complex scenes, this transition is particularly prone to boundary distortion and semantic ambiguity, as heterogeneous modalities exhibit inconsistent high-level feature distributions.

To address these challenges and ensure semantic coherence and geometric fidelity at high-level representations, we propose the SRAF module. SRAF adopts a refine-then-fuse strategy, in which modality-specific and potentially distorted semantic features are first rectified and stabilized before cross-modal fusion. This sequential design explicitly targets the dominant sources of high-level inconsistency in multimodal fusion, including distribution mismatch, semantic distortion, and boundary blur.

Specifically, SRAF refines high-level features through dual-dimensional attention modulation and frequency-aware semantic enhancement. In the first stage, channel–spatial recalibration mitigates misaligned feature distributions and suppresses unreliable semantic responses, enabling accurate semantic anchoring across modalities. Subsequently, residual semantic anchoring preserves low-frequency contextual information, stabilizing semantic alignment while preventing the loss of critical category cues. Building upon this stabilized representation, frequency-aware enhancement selectively restores boundary details and sharpens object contours, thereby improving the geometric integrity of high-level semantics.

Through this staged refinement process, SRAF effectively resolves semantic misalignment and boundary degradation, yielding semantically coherent and geometrically consistent high-level optical–SAR representations. The module consists of two core components: 1) channel–spatial semantic recalibration and 2) frequency-aware semantic enhancement, as illustrated in Figs. 4 and 5.

1) Channel–Spatial Semantic Recalibration: Given the high-level optical and SAR features $X_{\text{opt}}^{\text{high}}, X_{\text{sar}}^{\text{high}} \in \mathbb{R}^{C\times H/8\times W/8}$, each modality is first embedded via lightweight projections to normalize cross-modal feature distributions, laying the foundation for consistent attention operation

$$\widetilde{X}_{\text{opt}}^{\text{high}} = \varphi_{\text{opt}}\left(X_{\text{opt}}^{\text{high}}\right), \quad \widetilde{X}_{\text{sar}}^{\text{high}} = \varphi_{\text{sar}}\left(X_{\text{sar}}^{\text{high}}\right) \tag{9}$$

where $\varphi(\cdot)$ consists of a 1×1 convolution, BatchNorm, and ReLU, ensuring feature statistics alignment across modalities.

These embeddings normalize feature distributions across modalities, enabling consistent statistics for subsequent attention operations.

a) Channel modulation: Concatenated embeddings are passed through a shared channel attention mechanism to capture cross-modal semantic dependencies

$$F_{\text{cat}}^{\text{high}} = \text{Concat}\left[\widetilde{X}_{\text{opt}}^{\text{high}},\ \widetilde{X}_{\text{sar}}^{\text{high}}\right] \tag{10}$$

$$A_{\text{ch}} = \sigma\left(W_2 \cdot \delta\left(W_1 \cdot \text{GAP}\left(F_{\text{cat}}^{\text{high}}\right)\right)\right) \tag{11}$$

where $\text{GAP}(\cdot)$ denotes global average pooling and $\delta(\cdot)$ and $\sigma(\cdot)$ are ReLU and sigmoid activations, respectively. The resulting attention is decomposed into modality-specific channel maps, denoted as $A_{\text{ch,opt}}^{\text{high}}$ and $A_{\text{ch,sar}}^{\text{high}}$, which selectively emphasize semantically informative channels while suppressing redundant responses. In optical features, channel attention highlights reflectance-related cues and texture-related cues that are critical for discriminating planar materials such as vegetation and rooftops. In contrast, SAR channel attention prioritizes backscattering patterns associated with local and volumetric geometric structures, including slopes and building facades. By explicitly exploiting these modality-specific priors, the proposed channel attention mechanism enhances material discrimination in optical representations and preserves structural integrity in SAR features, thereby establishing a reliable semantic and geometric foundation for subsequent multimodal fusion.

b) Spatial modulation: Each modality's spatial attention is computed as

$$A_{\text{sp},m}^{\text{high}} = \sigma\left(\text{Conv}_{7\times 7}\text{Concat}\left[\text{AvgPool}\left(\widetilde{X}_m^{\text{high}}\right)\right.\right. \\ \left.\left.\text{MaxPool}\left(\widetilde{X}_m^{\text{high}}\right)\right]\right) \tag{12}$$

for $m \in \{\text{opt}, \text{sar}\}$. In the spatial dimension, optical attention primarily focuses on fine-grained textures and planar boundaries, such as farmland edges and wetland contours but remains sensitive to illumination variations that can induce boundary blurring under shadows. In contrast, SAR spatial attention highlights strong edges and structural continuity yet is susceptible to speckle-induced artifacts. By applying spatial attention in a modality-specific manner, SRAF enables complementary geometric correction. Optical spatial cues help suppress speckle-driven false edges in SAR features (e.g., pseudo-linear structures over bare ground), while SAR-derived contours stabilize blurred or fragmented boundaries in optical

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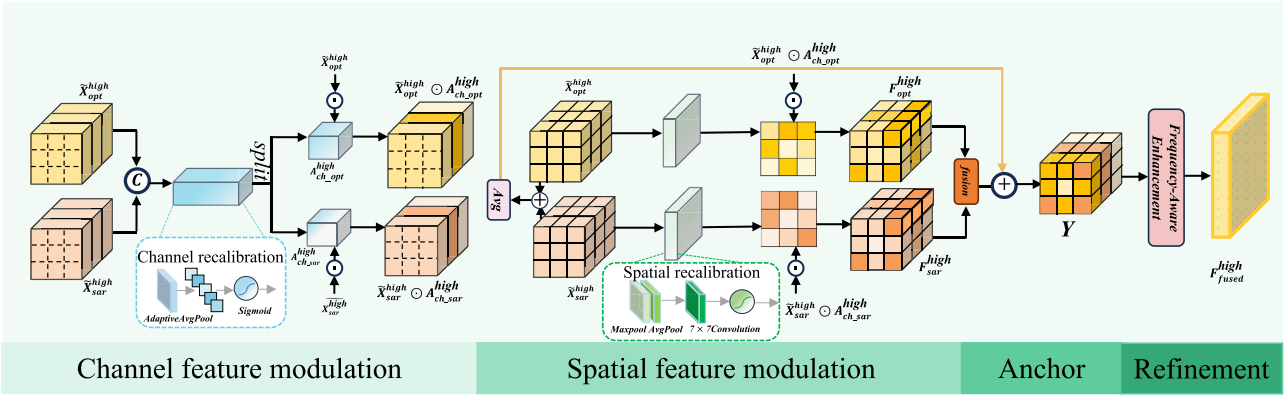


Fig. 4. Architecture of the SRAF module. The module performs modality-specific refinement through channel and spatial feature modulation, where channel recalibration captures cross-modal dependencies and spatial recalibration preserves structural localization. A residual anchor stabilizes semantic alignment across modalities.

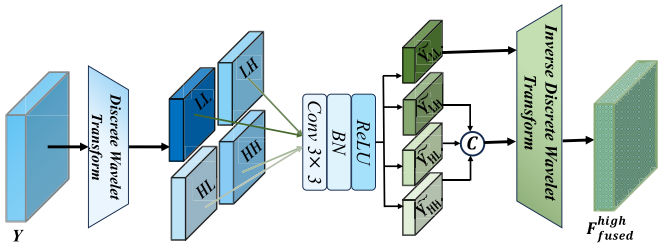


Fig. 5. Frequency-aware enhancement in the SRAF module. DWT decomposes fused semantics into low- and high-frequency components, followed by direction-aware reconstruction via inverse DWT, reinforcing boundary precision and cross-modal semantic consistency.

representations. This reciprocal spatial modulation effectively mitigates geometry inconsistency, boundary blur, and structural misalignment at the high semantic level. The recalibrated features are computed as

$$F_m^{\text{high}} = \widetilde{X}_m^{\text{high}} \odot A_{\text{ch},m}^{\text{high}} \odot A_{\text{sp},m}^{\text{high}} \tag{13}$$

where $\odot$ denotes the element-wise multiplication. As visualized in Fig. 4, SRAF disentangles modality-specific channel and spatial cues while preserving shared semantics.

c) Residual semantic anchoring: The refined optical and SAR features are concatenated and fused as

$$F_{\text{high}} = \psi\left(\text{Concat}\left[F_{\text{opt}}^{\text{high}},\ F_{\text{sar}}^{\text{high}}\right]\right). \tag{14}$$

To mitigate overmodulation and semantic distortion, SRAF introduces a residual semantic anchor

$$Y = F_{\text{high}} + \tfrac{1}{2}\left(X_{\text{opt}}^{\text{high}} + X_{\text{sar}}^{\text{high}}\right). \tag{15}$$

This residual connection retains low-frequency contextual cues from raw high-level features, stabilizing semantic alignment across modalities and preventing the loss of critical category information for small objects.

2) Frequency-Aware Enhancement: Finally, SRAF enhances structural and boundary fidelity using wavelet-based frequency decomposition (Fig. 5). Given Y, DWT decomposes the fused feature into one low-frequency component Y_L and three directional high-frequency sub-bands $\{Y_H(i)\}_{i=1}^3$

$$\{Y_{\text{H}}(i)\}_{i=1}^3 = \text{DWT}(Y). \tag{16}$$

Each branch is processed with orientation-specific convolutions to amplify valid structural signals and suppress high-frequency noise (e.g., residual speckle in SAR and cloud artifacts in optical)

$$\widetilde{Y}_L = F_L(Y_L), \quad \widetilde{Y}_H(i) = F_H(i)(Y_H(i)),\ i = 1,2,3 \tag{17}$$

and reconstructed via inverse DWT

$$F_{\text{fused}}^{\text{high}} = \text{IDWT}\left(\widetilde{Y}_L, \{\widetilde{Y}_H(i)\}_{i=1}^3\right). \tag{18}$$

This frequency-aware enhancement directly targets the dominant high-level challenge of boundary blur by strengthening directionally salient edges that are critical for maintaining geometric consistency, while preserving cross-modal semantic coherence for land-cover categories. The directional wavelet sub-bands are explicitly aligned with the geometric characteristics of key remote sensing targets: horizontal sub-bands enhance road continuity, vertical sub-bands reinforce riverbank boundaries, and diagonal sub-bands preserve building corner structures. Such orientation-specific enhancement enables targeted mitigation of boundary degradation and structural misalignment at the semantic level.

By jointly integrating dual-dimensional attention recalibration, residual semantic anchoring, and wavelet-based enhancement, the SRAF module transcends simple high-level feature concatenation and establishes a semantically consistent, structurally coherent, and modality-aware fusion mechanism. This integrated design effectively alleviates semantic drift and boundary misalignment that commonly arise in conventional optical–SAR high-level fusion approaches.

D. SACLF Module

To effectively integrate multimodal features across hierarchical levels while preserving directional structural cues and enhancing semantic–structural complementarity, we propose the SACLF module. As the final component in the proposed geometry-consistent fusion pipeline, SACLF bridges structure-rich low-level representations from the SDDF module and semantic-rich high-level representations from the SRAF module, addressing the intrinsic challenges of cross-level fusion in multimodal remote sensing scenarios. Specifically, SACLF

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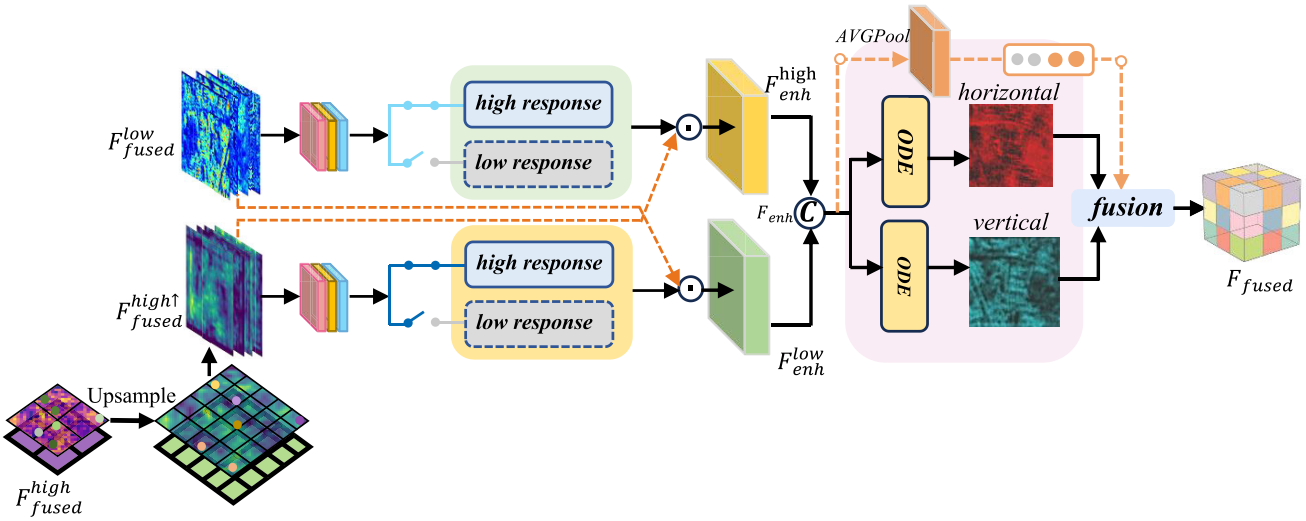


Fig. 6. Overview of the SACLF module. The module integrates bidirectional gating, orthogonal directional encoding, and global semantic context aggregation to enhance multimodal feature fusion. High-level semantics and low-level details are dynamically exchanged through adaptive gating, while orthogonal convolutions strengthen geometric consistency along horizontal and vertical orientations.

is designed to mitigate three key issues that arise during hierarchical fusion: 1) spatial resolution mismatch between high-level abstract semantics and low-level fine-grained edges; 2) progressive loss of directional geometric priors (e.g., road linearity and river orientation) during cross-scale feature interaction; and 3) insufficient scene-level contextual support for scattered or small-scale targets. As illustrated in Fig. 6, SACLF reconciles spatial resolution and semantic abstraction inconsistencies between high- and low-level representations, and explicitly encodes axis-aligned geometric priors to improve the representation of elongated and fine-grained targets.

Let $F_{\text{fused}}^{\text{high}} \in \mathbb{R}^{C_h \times H_h \times W_h}$ and $F_{\text{fused}}^{\text{low}} \in \mathbb{R}^{C_l \times H_l \times W_l}$ denote the high- and low-level fused features extracted from the optical–SAR branches, where $H_h < H_l$ and $W_h < W_l$ indicate spatial mismatch. To establish cross-scale consistency, the high-level feature is first upsampled to match the low-level resolution

$$F_{\text{fused}}^{\text{high}\uparrow} = \mathcal{U}\left(F_{\text{fused}}^{\text{high}}, (H_l, W_l)\right). \tag{19}$$

Both features are then projected into a unified latent semantic space of $C_m = \min(C_h, C_l)$ through cross-channel mapping functions

$$\widetilde{F}_{\text{high}}, \widetilde{F}_{\text{low}} = \phi_h\left(F_{\text{fused}}^{\text{high}\uparrow}\right), \ \phi_l\left(F_{\text{fused}}^{\text{low}}\right) \tag{20}$$

where $\phi_h(\cdot)$ and $\phi_l(\cdot)$ are implemented using a 1×1 convolution followed by batch normalization and ReLU activation to align semantic capacity and adapt channel scales. This projection plays a critical role in cross-level fusion for remote sensing imagery, as it unifies the semantic dimensionality of geometry-aligned low-level features from the SDDF module and semantically consistent high-level features from the SRAF module. By mitigating channel-scale mismatch, the projection ensures coherent interaction between structural and semantic representations across hierarchical levels.

Within this latent space, SACLF introduces a content-adaptive bidirectional gating mechanism to realize *high-guides-low* and *low-guides-high* enhancement

$$\left\{F_{\text{enh}}^{\text{high}}, F_{\text{enh}}^{\text{low}}\right\} = \left\{\widetilde{F}_{\text{high}} \odot \sigma\left(\widetilde{F}_{\text{low}}\right), \widetilde{F}_{\text{low}} \odot \sigma\left(\widetilde{F}_{\text{high}}\right)\right\} \tag{21}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function. Unlike one-way cross-level guidance, this bidirectional gating mechanism is specifically designed for remote sensing feature hierarchies. High-level gating injects category-level semantics from the SRAF module (e.g., vegetation versus water) into the geometry-aligned low-level features produced by SDDF, preventing fine-scale edges from being misinterpreted or overactivated. Conversely, low-level gating exploits reliable edge and shape cues to rectify erroneous high-level semantic activations, such as distinguishing narrow rivers from road shadows. Through this reciprocal interaction, SACLF achieves mutual correction between semantic abstraction and structural detail.

This soft, content-aware modulation enables effective bidirectional information flow, allowing high-level semantic context to guide fine spatial representations while simultaneously refining semantic responses using detailed geometric cues.

To strengthen the representation of elongated and directionally structured targets, the enhanced features $F_{\text{enh}}^{\text{high}}$ and $F_{\text{enh}}^{\text{low}}$ are concatenated and fused through a 1×1 convolution

$$F_{\text{enh}} = \text{Conv}_{1\times1}\left(\text{Concat}\left[F_{\text{enh}}^{\text{high}}, F_{\text{enh}}^{\text{low}}\right]\right). \tag{22}$$

An orthogonal directional encoding (ODE) is then applied along horizontal and vertical axes

$$\begin{aligned} F_h, F_v &= \text{ODE}\left(F_{\text{enh}}; k_h = 1, k_w = 5\right) \\ &\text{ODE}\left(F_{\text{enh}}; k_h = 5, k_w = 1\right). \end{aligned} \tag{23}$$

This encoding is explicitly matched to the geometric properties of key remote sensing targets: horizontal kernels enhance the

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linear extension of roads and farmland ridges, while vertical kernels reinforce the boundary integrity of riverbanks and building facades. These orthogonal depthwise convolutions produce independent response flows along horizontal and vertical orientations, explicitly encoding directional constraints that enhance the network's sensitivity to linear and anisotropic structures. Moreover, a global context path aggregates scene-level priors through average pooling

$$F_g = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{AvgPool}\left(F_{\mathrm{enh}}\right)\right)\right)\right)$$
$$F_{\mathrm{fused}} = \mathrm{Concat}\left(F_h, F_v, F_g\right). \tag{24}$$

This combination of directional encoding and global aggregation yields a geometry-consistent and structure-aware cross-level representation that effectively bridges semantic abstraction and structural detail for robust multimodal segmentation.

Notably, SACLF forms a complete geometric-consistency loop with the preceding SDDF (low-level geometric alignment) and SRAF (high-level semantic calibration) modules. SDDF lays the structural foundation, SRAF elevates semantic coherence, and SACLF unifies cross-level semantics and structure, realizing end-to-end geometry-preserving multimodal fusion for remote sensing segmentation. By explicitly encoding cross-scale geometric continuity, SACLF ensures that elongated and anisotropic land-cover structures—such as roads, rivers, and linear infrastructures—are consistently represented across abstraction levels. This property is particularly critical for geoscience applications where topological integrity, rather than pixel-wise accuracy alone, determines the usability of segmentation outputs.

E. Multiobjective Adaptive Loss for Geometry-Consistent Optimization

To further enhance training stability and convergence, we design a composite loss function that jointly optimizes semantic accuracy and geometric fidelity. The total objective integrates multiple complementary components to balance the contributions of different tasks

$$L_{\mathrm{total}} = \alpha_1 L_{\mathrm{CE}} + \alpha_2 L_{\mathrm{Dice}} + \alpha_3 L_{\mathrm{Focal}} + \alpha_4 L_{\mathrm{Boundary}} \tag{25}$$

where L_{CE}, L_{Dice}, L_{Focal}, and L_{Boundary} address class-imbalance, small-object segmentation, hard-to-classify regions, and precise boundary delineation, respectively. Their mathematical formulations are defined as

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log\left(\hat{y}_i\right) \tag{26}$$

$$L_{\mathrm{Dice}} = 1 - \frac{2\sum_i y_i \hat{y}_i}{\sum_i y_i^2 + \sum_i \hat{y}_i^2} \tag{27}$$

$$L_{\mathrm{Focal}} = -\frac{1}{N}\sum_{i} \left(1-\hat{y}_i\right)^{\gamma} y_i \log\left(\hat{y}_i\right) \tag{28}$$

$$L_{\mathrm{Boundary}} = \frac{1}{N}\sum_{i} \left\|\nabla y_i - \nabla \hat{y}_i\right\|_1 \tag{29}$$

where y_i and $\hat{y}_i$ denote the ground truth and predicted probability for pixel i, γ is the focusing parameter in the focal loss, and ∇ represents the gradient operator used to compute boundary discrepancies. The combination of these losses ensures that the network learns semantically accurate predictions while maintaining local geometric continuity and sharp object boundaries, which are essential for geometry-consistent multimodal segmentation. To adaptively adjust the relative importance of each loss term during training, we employ an adaptive weighting strategy [48], [49] based on the recent performance of each component

$$\alpha_i(k) = \frac{e^{\beta s_i(k)}}{\sum_{j=1}^{4} e^{\beta s_j(k)}} \tag{30}$$

where $s_i(k) = L_i(k) - L_i(k-1)$ represents the rate of change of the ith loss term (positive values indicate performance degradation and negative values indicate improvement). β is a temperature scaling hyperparameter that controls the sensitivity of weight adjustments, with a larger value of β amplifying the weight differences to emphasize underoptimized components, while smaller values ensure more uniform weighting. We set $\beta = 0.1$ as the initial value [48], following SoftAdapt's validated default, to strike a balance between sensitivity and stability.

During training, the mechanism first computes $s_i(k)$ for each loss term, scales it by β, maps the scaled values to non-negative weights using the exponential function, and normalizes them via softmax (ensuring the sum of weights equals 1). Ultimately, higher weights $alpha_i(k)$ are assigned to underoptimized components [those with positive $s_i(k)$], while lower weights are assigned to converged components [those with negative $s_i(k)$]. This mechanism enables the model to automatically balance gradient propagation across different loss components, progressively aligning semantic and geometric features throughout training.

IV. Experiments and Analysis

A. Dataset Description

1) WHU-OPT-SAR Dataset: This dataset is a large-scale benchmark for multimodal land-use segmentation. It covers 51 448.56 km^2 in Hubei, China, with 100 coregistered optical (RGBN) and SAR image pairs. Optical images from GaoFen-1 (2 m) were resampled to 5 m to match SAR images from GaoFen-3 [50]. Each pair has a size of 5556×3704 pixels and pixel-level labels for seven classes: farmland, forest, city, village, water, road, and others. Following the hold-out strategy, 80 pairs are used for training, 10 pairs are used for testing, and 10 pairs are used for validation, with each image divided into 256×256 patches, yielding 17 640 training, 5880 validation, and 5880 testing samples. By combining spectral–spatial cues from optical images with structural information from SAR, the dataset provides a challenging benchmark for multimodal semantic segmentation [50].

2) YESeg-OPT-SAR Dataset: This dataset is a high-resolution multimodal benchmark for land-use classification, combining optical and SAR imagery. It contains 2231 coregistered image pairs, with optical images sourced from Google Earth L18 products and SAR images acquired via satellite. The optical images have a spatial resolution of 0.5 m, while

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the SAR images also have a 0.5-m resolution [51]. Each image pair has a size of 256 × 256 pixels, with pixel-level annotations across seven land-use classes: background, bare ground, vegetation, trees, houses, water, road, and others [51]. The dataset is split into training and testing subsets, with each image pair annotated for supervised learning tasks. The dataset is randomly split into 80% training and 20% testing sets to ensure balanced spatial distribution and consistent land-cover proportions across subsets [51]. By leveraging both spectral–spatial information from optical images and structural features from SAR, the YESeg-OPT-SAR dataset offers a valuable benchmark for advancing multimodal semantic segmentation in remote sensing.

3) Pohang Dataset: To validate the adaptability of StructFuse-Net to heterogeneous optical–SAR data and weather-disturbed scenarios, we use the Pohang Dataset [52], which consists of GF-2 optical and GF-3 SAR imagery covering coastal, mountainous, and industrial areas in South Korea. The original image size is 10 240×9216 pixels, cropped into 256 × 256 patches with a stride of 128, resulting in 3087 training and 3086 testing patches. Notably, the dataset includes a cloud-disturbed subset [52], where fog-like clouds are artificially added using GIMP's fog filter, simulating optical degradation caused by light fog or haze. This challenges geometric consistency in multimodal fusion.

B. Experimental Settings

1) Implementation Details: All experiments were conducted on a single NVIDIA GeForce RTX 3090 GPU using the PyTorch framework. To facilitate stable training, the proposed model was optimized using AdamW with an initial learning rate of 1×10^{-3}, a weight decay of 3×10^{-4}, and betas of (0.9, 0.999). Following a warmup strategy, the learning rate was gradually increased to the initial value within the first five epochs. The learning rate was then scheduled using a cosine annealing strategy with a minimum value of 1×10^{-7}. Training was conducted for 100 epochs with a batch size of 8, and a fixed random seed of 42 was used to ensure reproducibility. During training, standard data augmentation techniques were applied, including random horizontal and vertical flipping, random rotation within $\pm 10°$, and random scaling in the range of 0.9–1.1. These augmentations improve generalization while avoiding overly aggressive transformations. For evaluation, overlapping sliding windows were used to generate prediction patches. The predictions in overlapping regions were averaged to reduce boundary errors. Multiscale augmentation with scales of [0.9, 1.0, 1.1] and random flipping were applied during evaluation to further enhance prediction stability, while keeping inference time within an acceptable range.

2) Evaluation Metrics: To assess the performance of our method, we use overall accuracy (OA), Kappa coefficient, mean intersection over union (mIoU), and per-class accuracy. OA measures the proportion of correctly classified pixels, while Kappa evaluates agreement with ground truth. mIoU computes the average intersection over union across classes, and per-class accuracy reports performance for each individual class. These metrics provide a comprehensive evaluation of both overall and class-specific segmentation accuracy.

3) Comparative Models: To comprehensively evaluate the effectiveness and generalization of the proposed StructFuse-Net, we compared it with seven SOTA methods on WHU-OPT-SAR and YESeg-OPT-SAR datasets through both quantitative and qualitative analyses. The selected methods cover a wide range of representative paradigms, including CNN-based, attention-based, transformer-based, and state-space architectures.

1) *Deeplabv3 +:* Chen et al. [41] enhanced the original DeepLabv3 by introducing an efficient decoder to refine segmentation boundaries. They further adopted depthwise separable convolutions in the atrous spatial pyramid pooling (ASPP) module, significantly improving multiscale feature representation while reducing computational complexity.
2) *SA-Gate:* Chen et al. [23] proposed a separation-and-aggregation gate architecture for multimodal semantic segmentation. The gate module adaptively filters and recalibrates features from each modality prior to fusion, enabling more discriminative and noise-robust feature learning.
3) *CMFNet:* Ma et al. [34] designed a cross-modal multiscale fusion network that integrates a cross-modal transformer (CMTrans) and a multiscale context-augmented transformer (MCATrans). It captures long-range dependencies and global context while employing a combined squeeze-and-excitation (CSE) module for efficient feature integration.
4) *CMGFNet:* Hosseinpour et al. [24] developed CMGFNet with dual encoders to extract features from VHR and DSM data. A GFM and a top-down semantic refinement strategy are used to combine multilevel cues, improving building delineation and class boundary accuracy.
5) *MCANet:* Li et al. [50] introduced a multimodal cross-attention network for optical–SAR land-use classification. Its design employs modality-specific feature extractors and a multiscale attention fusion module, effectively leveraging the complementary properties of both modalities.
6) *RS3Mamba:* Ma et al. [35] proposed RS3Mamba, a dual-branch remote sensing segmentation network that integrates a VSS auxiliary branch with a convolutional main branch to efficiently model long-range dependencies. A collaborative completion module (CCM) further refines and fuses multilevel features, leading to strong performance across multiple benchmarks.
7) *FTransUNet:* Ma et al. [37] designed a multilevel fusion framework combining CNNs and transformers. FTransUNet employs shallow-level feature fusion (SFF) blocks and a fusion vision transformer (FVit) equipped with adaptively mutually boosted attention (Ada-MBA) layers, achieving superior segmentation accuracy on multimodal benchmarks.

C. Experimental Results

1) Results on the WHU-OPT-SAR Dataset: To comprehensively evaluate the performance of the proposed StructFuse-Net in multimodal RSI segmentation, extensive experiments

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TABLE I

QUANTITATIVE RESULTS OF VARIOUS METHODS ON THE WHU-OPT-SAR DATASET, INCLUDING PER-CLASS AND OVERALL METRICS (OA, KAPPA, AND MIOU). STRUCTFUSE-NET ACHIEVES THE BEST OVERALL PERFORMANCE ACROSS MOST CATEGORIES

Method	OA(%)							OA(%)	Kappa(%)	mIoU(%)
	agriculture	city	village	water	forest	road	others			
Deeplabv3+	70.2	65.9	60.2	62.1	87.6	51.7	9.3	77.4	63.7	51.2
SA-Gate	69.4	68.7	61.4	53.4	87.4	50.7	6.1	76.1	63.3	50.9
MCANet	73.1	70.1	56.8	63.2	89.1	56.1	9.1	78.1	68.5	51.9
CMFNet	76.2	69.3	63.1	61.3	87.4	55.3	8.4	77.5	65.2	51.5
CMGFNet	75.7	65.7	60.1	62.5	89.3	56.6	10.2	77.9	66.3	51.3
RS3Mamba	70.9	63.9	60.7	62.6	90.1	56.7	9.4	79.1	69.3	52.4
FTransUNet	**79.3**	70.6	66.7	65.9	**94.2**	57.3	13.8	80.5	69.7	52.7
StructFuse-Net(ours)	77.1	**72.5**	**67.4**	**68.9**	90.4	**65.1**	**26.8**	**82.7**	**71.4**	**54.5**

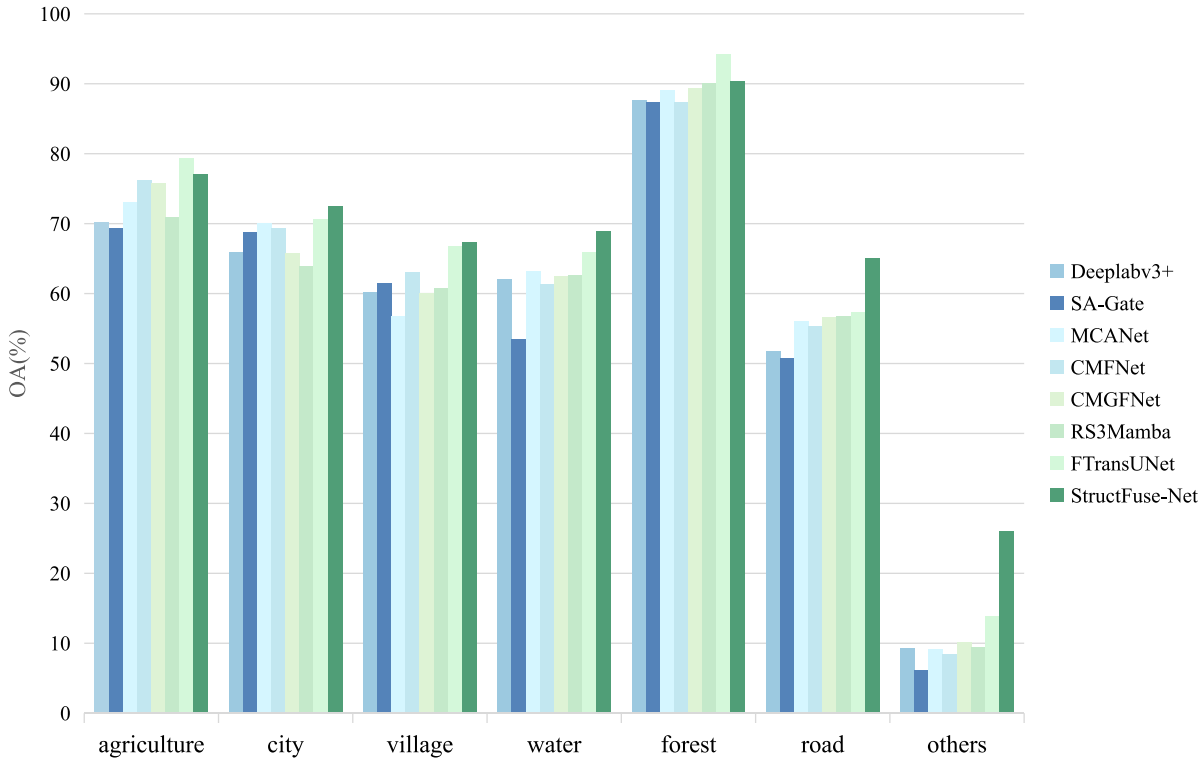


Fig. 7. Per-class OA comparison on the WHU-OPT-SAR dataset. StructFuse-Net consistently achieves the highest accuracy across all land-cover categories, particularly in complex classes such as roads, water, and villages, highlighting its robustness under environmental interference.

were conducted on the WHU-OPT-SAR dataset. The quantitative results are summarized in Table I, including OA, Kappa coefficient, mean mIoU, and per-category accuracies. The class-wise OA distribution is further illustrated in Fig. 7, while representative qualitative comparisons are presented in Fig. 8.

StructFuse-Net consistently achieves the best performance across all competing approaches, with an OA of 82.7%, a Kappa coefficient of 71.4%, and an mIoU of 54.5%. As shown in Table I, our method improves overall segmentation accuracy by more than 2% compared with the strongest baselines such as FTransUNet and RS3Mamba, confirming the model's strong cross-modal representation and generalization ability. In Fig. 7, StructFuse-Net maintains leading accuracy across all land-cover categories, particularly excelling in complex classes such as road, water, and village, where structural consistency and semantic discrimination are challenging.

Qualitative comparisons in Fig. 8 further highlight the advantages of our model under challenging imaging conditions. In mixed landscapes where water and roads intersect, optical images [Fig. 8(a1)] offer fine texture but suffer from weather-induced blurring, while SAR images [Fig. 8(b1)] provide structural integrity but are contaminated by speckle noise. Benefiting from the SDDF module with bidirectional cross-modal attention, StructFuse-Net [Fig. 8(k1)] effectively aligns optical edges with SAR directional structures,

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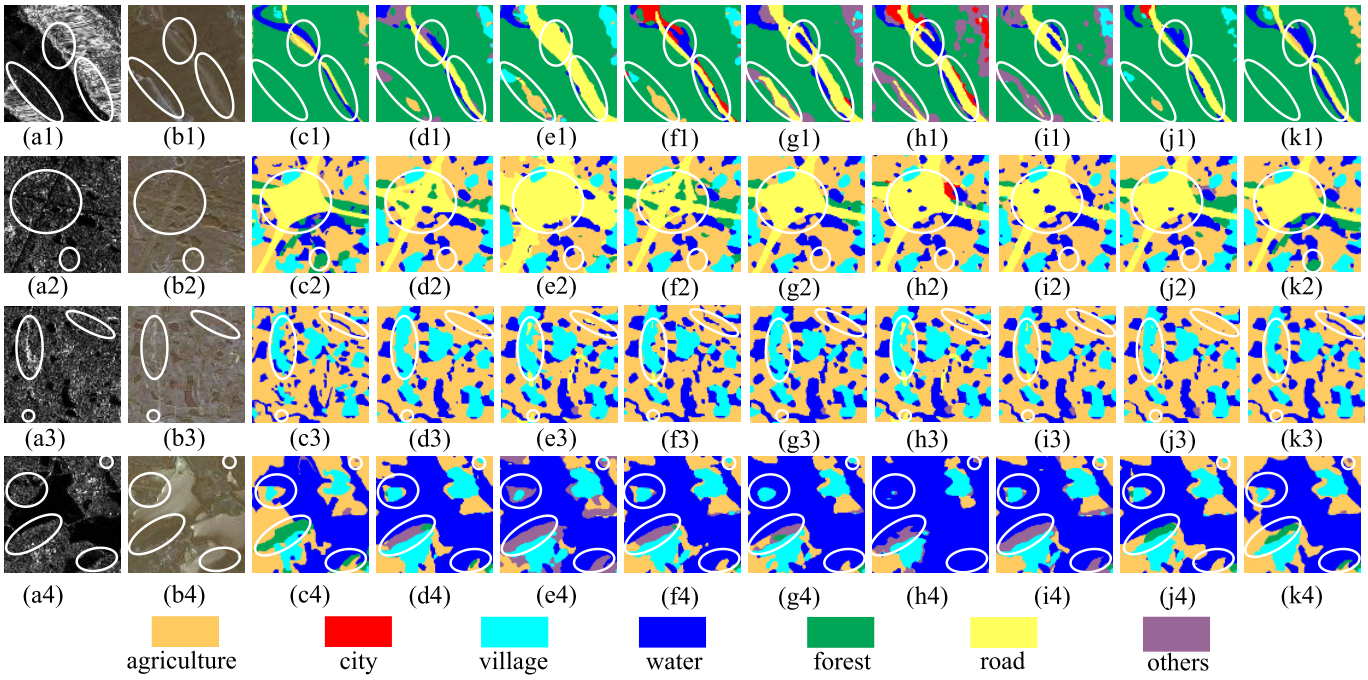


Fig. 8. Visual comparison of segmentation results on the WHU-OPT-SAR dataset. (a1)–(a4) Optical images. (b1)–(b4) SAR images. (c1)–(c4) Ground truth. (d1)–(k4) Predictions of DeepLabv3 +, SA-Gate, MCANet, CMFNet, CMGFNet, RS3Mamba, FTransUNet, and StructFuse-Net. StructFuse-Net produces clearer boundaries and structurally consistent segmentations across diverse land-cover types.

suppressing modality-specific noise and improving water segmentation accuracy by over 15%.

In scenes involving intricate road networks such as intersections and roundabouts [Fig. 8(d2)–(k2)], most comparative methods exhibit incomplete boundaries and discontinuities. The proposed SACLF module explicitly models geometric priors through orthogonal directional encoding, enabling the network to preserve fine linear features and achieve an OA of 65.1% in road segmentation. In agricultural and settlement areas containing farmland, villages, and water bodies, StructFuse-Net demonstrates strong cross-category discrimination. The SRAF module's semantic refinement mechanism ensures the alignment of modality-specific cues, enabling accurate delineation of farmland (77.1%) and village (67.4%) regions.

In regions degraded by cloud occlusion or specular reflection [Fig. 8(b3) and (b4)], optical imagery experiences severe spectral distortion, often leading to large-scale misclassification. Through frequency-aware multiscale fusion and adaptive modulation, StructFuse-Net effectively mitigates environmental interference, achieving OA values of 68.9% and 67.4% for water and village categories, respectively. Furthermore, in vegetation–settlement overlapping areas [Fig. 8(b4)], where optical images suffer from shadow effects and SAR data exhibit volumetric scattering, the structure-aware fusion strategy of SACLF enables the model to distinguish between forests and built-up regions with higher precision, reducing class confusion and boundary ambiguity.

Overall, these results verify the effectiveness of our structure-aware multimodal fusion framework. The collaborative integration of the SDDF, SRAF, and SACLF modules allows StructFuse-Net to jointly enhance semantic and structural representations, overcoming key challenges such as noise interference, cross-modal inconsistency, and ambiguous category boundaries. Both quantitative and qualitative results confirm that StructFuse-Net achieves SOTA performance on the WHU-OPT-SAR dataset, providing a robust and generalizable solution for fine-grained multimodal RSI interpretation.

2) Results on the YESeg-OPT-SAR Dataset: To further validate the generalization capability of the proposed StructFuse-Net, we conducted comparative experiments on the YESeg-OPT-SAR dataset against seven representative multimodal segmentation models. The quantitative results are summarized in Table II, while representative qualitative comparisons are shown in Fig. 9.

StructFuse-Net achieves an OA of 81.4%, a Kappa coefficient of 74.7%, and an mIoU of 62.4%, surpassing the best baselines by 1.3%–3.5% in OA and 2.1%–5.0% in mIoU. Notably, the network achieves significant gains for challenging categories such as *other* (45.5%, + 2.0%), *roads* (80.8%, + 1.7%), and *water* (83.7%, + 1.3%), indicating its capability to balance both dominant and minority classes. This overall improvement demonstrates that StructFuse-Net effectively mitigates the common performance bias toward dominant land-cover types observed in multimodal segmentation networks.

These quantitative improvements stem from the network's structure-aware fusion mechanism. At the shallow level, the SDDF module preserves modality-specific geometric cues—retaining optical edge textures while suppressing SAR-induced speckle noise—thereby producing geometrically consistent feature representations. At higher semantic levels,

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TABLE II

QUANTITATIVE RESULTS OF VARIOUS METHODS ON THE YESEG-OPT-SAR DATASET, INCLUDING PER-CLASS AND OVERALL METRICS (OA, KAPPA, AND MIOU). STRUCTFUSE-NET CONSISTENTLY OUTPERFORMS BASELINES IN OVERALL AND MOST CATEGORY-LEVEL ACCURACIES

Method	OA(%)							OA(%)	Kappa(%)	mIoU(%)
	bare ground	low vegetation	trees	houses	water	roads	other			
Deeplabv3+	73.7	79.1	60.2	68.8	72.8	71.3	27.9	75.3	66.6	56.9
SA-Gate	75.2	80.1	59.8	69.3	78.2	73.1	28.7	76.9	68.5	59.2
MCANet	76.5	80.3	60.9	73.2	75.2	74.8	40.1	77.3	69.2	60.3
CMFNet	78.5	82.4	63.9	68.6	75.1	76.2	26.1	78.8	71.1	56.3
CMGFNet	79.1	82.8	67.2	73.1	81.9	78.8	24.3	80.1	72.8	57.9
RS3Mamba	78.7	82.6	64.1	75.3	81.3	78.3	27.1	79.6	72.3	57
FTransUNet	80.2	83.1	**69.9**	75.6	76.3	78.7	33.5	79.7	72.4	57.5
StructFuse-Net(ours)	**80.5**	**84.2**	68.7	**79.1**	**83.7**	**80.8**	**45.5**	**81.4**	**74.7**	**62.4**

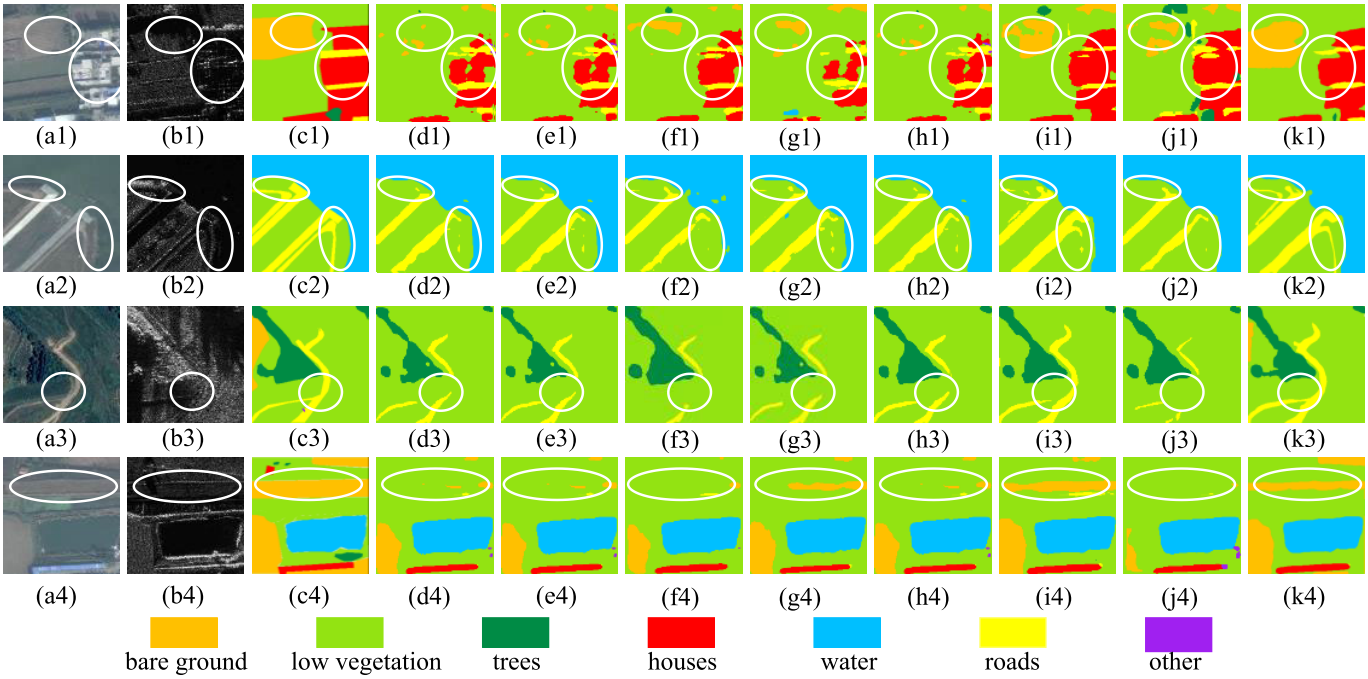


Fig. 9. Visual comparison of segmentation results on the YESeg-OPT-SAR dataset. (a1)–(a4) Optical images. (b1)–(b4) SAR images. (c1)–(c4) Ground truth. (d1)–(k4) Predictions from DeepLabv3 +, SA-Gate, MCANet, CMFNet, CMGFNet, RS3Mamba, FTransUNet, and StructFuse-Net. StructFuse-Net demonstrates superior delineation of heterogeneous categories such as *other, roads*, and *water*, maintaining structural continuity and suppressing noise in challenging scenes.

the SRAF module ensures cross-modal alignment and semantic coherence by jointly refining spatial and channel-level activations, enabling texture-rich optical information and structure-oriented SAR signals to converge into semantically consistent, boundary-accurate features. Meanwhile, the SACLF module bridges hierarchical layers through direction-sensitive gating and orthogonal encoding, enhancing the representation of elongated and irregular objects while maintaining overall semantic consistency. This cooperative design results in precise delineation of both homogeneous and geometrically complex regions.

The qualitative results in Fig. 9 further validate the robustness of the proposed framework across diverse environments. In urban scenes [see Fig. 9(k1)], StructFuse-Net effectively distinguishes roads from adjacent buildings and maintains the topological continuity of elongated structures. For rural and riverine regions [Fig. 9(k2) and (k3)], it delivers well-defined water and road boundaries with reduced speckle interference, outperforming competing models that often exhibit fragmented or noisy predictions. Even in visually heterogeneous regions [Fig. 9(a4)–(k4)], where the *other* category consists of sparsely distributed targets with unstable spectral properties, StructFuse-Net achieves significantly higher accuracy by combining optical semantics with SAR structural priors.

Overall, the superior performance of StructFuse-Net on the YESeg-OPT-SAR dataset highlights its robustness and

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TABLE III

QUANTITATIVE RESULTS OF VARIOUS METHODS ON THE POHANG DATASET, INCLUDING PER-CLASS AND OVERALL METRICS (OA, KAPPA, AND MIOU). STRUCTFUSE-NET ACHIEVES THE BEST OVERALL PERFORMANCE ACROSS MOST CATEGORIES

Method	OA(%)					OA(%)	Kappa(%)	mIoU(%)
	Building	Greenery	Farmland	Water	Road			
Deeplabv3+	91.7	83.8	87.6	98.2	73.2	90.3	79.2	76.9
SA-Gate	89.3	74.2	90.3	96.5	88.6	87.9	82.1	79.2
MCANet	92.6	86.9	93.2	98.2	69.8	92.1	81.4	79.3
CMFNet	93.2	84.1	91.2	98.7	89.3	91.0	85.6	82.5
CMGFNet	92.2	83.5	90.7	98.5	88.4	90.6	84.8	81.7
RS3Mamba	93.5	85.2	91.8	98.2	89.5	91.7	83.1	83.2
FTransUNet	**96.8**	88.5	86.1	98.2	70.2	92.6	86.7	84.2
StructFuse-Net(ours)	95.6	**90.3**	**94.7**	**99.6**	**92.1**	**94.8**	**89.3**	**86.5**

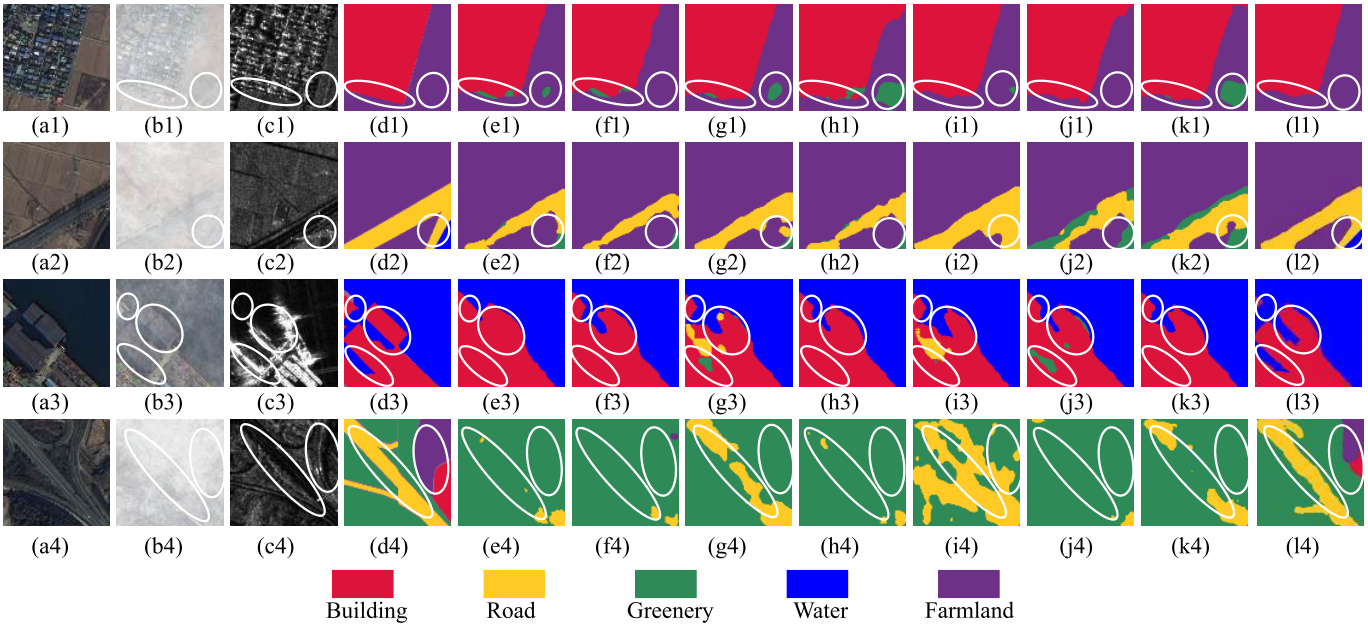


Fig. 10. Visual comparison of segmentation results on the Pohang dataset. (a1)–(a4) Origin images. (b1)–(b4) Optical images. (c1)–(c4) SAR images. (d1)–(d4) Ground truth. (e1)–(l4) Predictions from DeepLabv3 +, SA-Gate, MCANet, CMFNet, CMGFNet, RS3Mamba, FTransUNet, and StructFuse-Net. StructFuse-Net demonstrates superior performance in delineating heterogeneous land-cover categories, such as roads, water, and building while preserving the structural continuity of linear features, including *roads, water*, and *buildings*. It also effectively suppresses thin cloud-induced noise in challenging scenes.

generalization to varying imaging conditions and land-cover complexities. The synergistic operation of the SDDF, SRAF, and SACLF modules ensures geometry-consistent low-level guidance, semantically aligned high-level representation, and cross-level structural coherence. Consequently, StructFuse-Net achieves quantitatively superior and visually consistent segmentation across a wide range of multimodal remote sensing scenarios.

3) Results on the Pohang Dataset: To evaluate the robustness of StructFuse-Net under cloud-induced optical degradation, we conducted experiments on the cloud-disturbed subset of the Pohang dataset. The quantitative results are summarized in Table III. StructFuse-Net achieves an OA of 94.8%, a Kappa coefficient of 89.3%, and an mIoU of 86.5%, outperforming competing methods across most categories. Notably, it shows clear advantages in cloud-sensitive classes such as road (92.1%, + 2.6%), greenery (90.3%, + 1.8%), and water (99.6%, + 1.1%) compared with the strongest baselines.

The performance gains can be attributed to the proposed structure-aware fusion strategy, which effectively mitigates the adverse impact of cloud-induced optical degradation. In particular, the model leverages stable structural cues from SAR imagery to compensate for degraded optical textures, leading to improved boundary localization and geometric consistency. This effect is especially pronounced for elongated and structure-sensitive objects, such as roads and rivers, whose spatial continuity is often disrupted under heavy cloud coverage.

Qualitative comparisons in Fig. 10 further corroborate these observations. In heavily cloud-obscured scenes [see Fig. 10(b1)–(b4)], StructFuse-Net produces coherent and accurate segmentation maps, whereas representative baselines, such as Deeplabv3 + and MCANet, exhibit fragmented

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TABLE IV
ABLATION STUDY OF INDIVIDUAL MODULES ON THE WHU-OPT-SAR DATASET

SDDF	SRAF	SACLF	OA(%)	mIoU(%)
✓			80.3	51.8
✓	✓		81.6	53.2
✓	✓	✓	**82.7**	**54.5**

predictions or misclassification. For cloud-covered roads and vegetation regions [Fig. 10(e1)–(l4)], StructFuse-Net maintains clearer boundaries and more consistent structures, demonstrating stronger robustness under severe optical degradation.

Overall, the results on the Pohang dataset indicate that StructFuse-Net maintains stable performance under challenging sensing conditions, highlighting its robustness to optical degradation and its potential for practical deployment in real-world multimodal remote sensing applications.

D. Ablation Studies and Component Analysis

1) Effectiveness of Individual Modules: To assess the individual contributions of the proposed modules, we conducted ablation experiments on the WHU-OPT-SAR dataset, as summarized in Table IV. Incorporating the SDDF module improves global geometric perception, achieving an OA of 80.3% and an mIoU of 51.8%. This enhancement primarily benefits structured targets such as rivers and roads, where directional cues are crucial for geometry preservation. When the SRAF module is further introduced, the model becomes more responsive to fine-scale local variations, resulting in improved recognition of small and texture-rich objects and an mIoU increase to 53.2%. Finally, the inclusion of the SACLF module enables hierarchical interaction between global and local representations, maintaining contour continuity and spatial coherence in complex scenes. The full StructFuse-Net achieves the highest performance with an OA of 82.7% and an mIoU of 54.5%, demonstrating that the three modules provide complementary and synergistic effects for multimodal segmentation.

To further analyze the functional behavior of each component, Fig. 11 visualizes the feature activation maps under representative scenes. The SDDF module produces continuous activations along linear and edge-like structures, reflecting its strong ability to capture large-scale geometric patterns. However, it remains less effective in handling complex textures or gradual transitions, which occasionally results in boundary discontinuities. Complementarily, the SRAF module exhibits stronger responses to local salient regions, effectively enhancing fine-scale textures such as urban buildings or narrow river branches. Yet, in the absence of global structural constraints, its activations appear fragmented and less consistent across categories.

In contrast, the SACLF module integrates the advantages of both perspectives, yielding coherent and geometry-consistent responses. Its heatmaps exhibit continuous contours and well-differentiated regions, particularly along object boundaries and multiclass intersections. This demonstrates that SACLF not only reinforces spatial continuity but also preserves geometric fidelity across modalities.

Overall, the three modules collectively contribute to a progressive enhancement. SDDF focuses on global geometric awareness, SRAF emphasizes local saliency, and SACLF harmonizes both through cross-level feature fusion. These results visually and quantitatively validate the complementary design of StructFuse-Net and its effectiveness in achieving geometry-consistent optical–SAR segmentation.

2) Effectiveness of Wavelet Selection in StructFuse-Net: Wavelet-based decomposition is a key operation in StructFuse-Net, contributing to both low-level structural enhancement and high-level semantic refinement. To systematically assess the impact of different wavelet bases, we conducted comparative experiments in both the SDDF and SRAF modules using four common wavelets: db1, db4, sym4, and coif1.

a) Effectiveness of wavelets in the SDDF module: To further verify the effectiveness of wavelet selection for low-level feature enhancement, we compared four commonly used wavelet bases—db1, db4, sym4, and coif1—within the SDDF module of StructFuse-Net. The quantitative results in Table V and the corresponding heatmap visualizations in Fig. 12 jointly demonstrate that the db1 wavelet yields the best overall performance, achieving the highest OA of 82.75% and mIoU of 54.5%.

Across four representative optical–SAR scenes, db1 produces clearer and more continuous responses along linear structures such as rivers, roads, and urban boundaries. In contrast, db4 and sym4 result in slightly blurred edge representations, particularly around narrow rivers and sharp urban transitions, while coif1 captures general texture patterns but lacks directional sensitivity to elongated features. This observation suggests that db1's short support length and compactness allow it to more effectively enhance local geometric cues and suppress irrelevant noise components.

Overall, db1 preserves local geometric integrity and boundary sharpness in low-level representations, thereby providing a robust foundation for subsequent high-level feature fusion and structure-aware processing. These findings confirm that appropriate wavelet selection in the SDDF module plays a crucial role in maintaining geometric continuity and promoting geometry-consistent multimodal segmentation.

b) Effectiveness of wavelets in the SRAF module: In the high-level semantic enhancement stage, the DWT is employed in the SRAF module to perform multiscale decomposition, enabling a comparative analysis of different wavelet bases. As summarized in Table VI and illustrated in Fig. 13, the db1 wavelet achieves the highest classification accuracy of 82.75% and an mIoU of 54.5%, outperforming db4, sym4, and coif1. Heatmap visualizations show that db1 produces continuous and coherent activations along linear object boundaries while maintaining the geometric continuity of structured categories such as roads and river systems. Moreover, it effectively enhances the activation of sparse and small-scale targets, including isolated buildings and minor water bodies.

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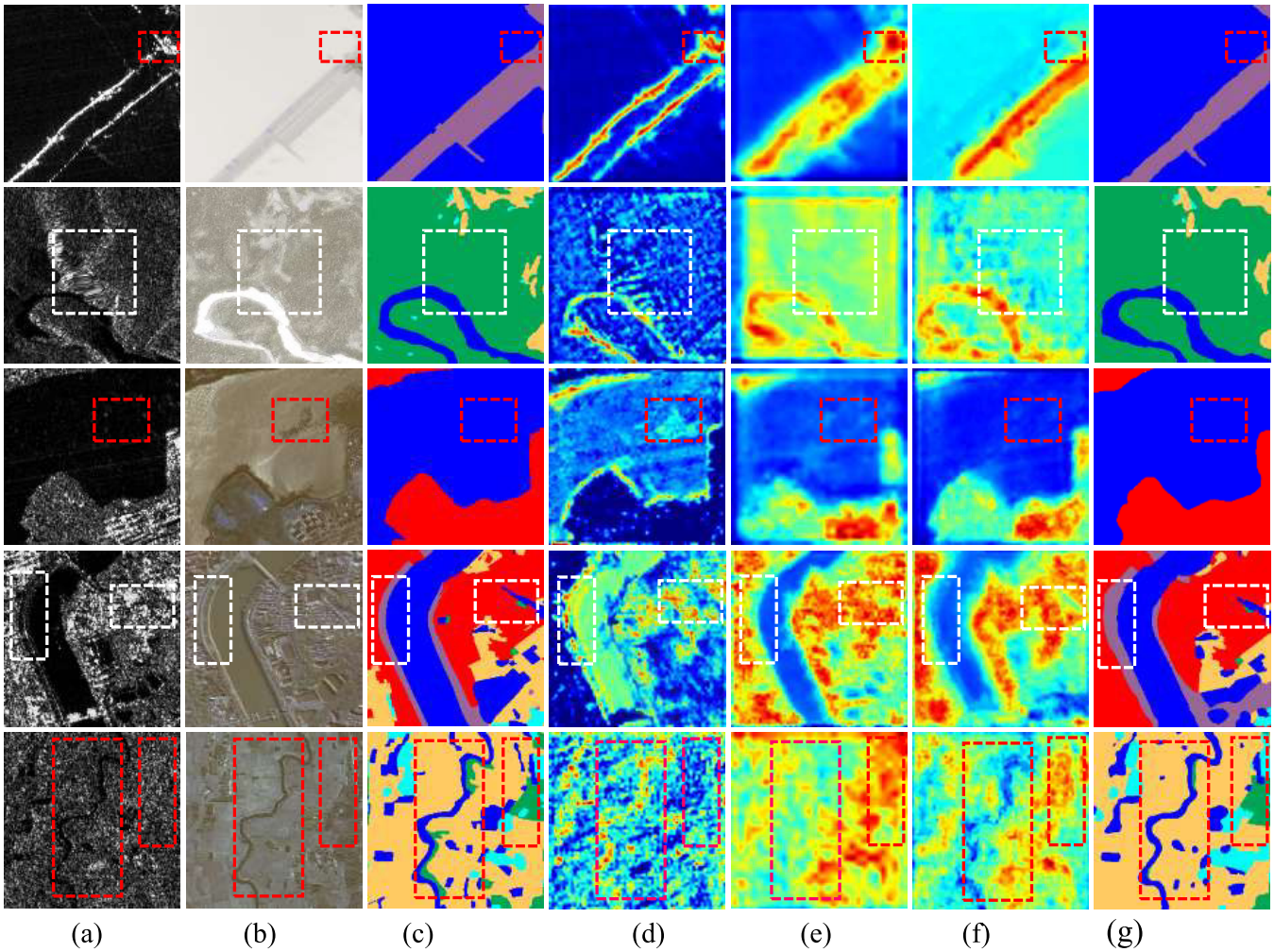


Fig. 11. Heatmap visualizations of different module configurations on representative optical–SAR scenes. (a) SAR, (b) optical, (c) ground truth, (d) SDDF, (e) SRAF, (f) SACLF, and (g) StructFuse-Net. SDDF primarily highlights large-scale geometric structures such as rivers and roads, while SRAF captures fine-grained local saliency and texture details. SACLF integrates both global and local cues, preserving spatial continuity and clear class boundaries. The full StructFuse-Net achieves geometry-consistent segmentation across multimodal inputs.

TABLE V
COMPARISON OF DIFFERENT WAVELET BASES USED FOR LOW-LEVEL ENHANCEMENT IN THE SDDF MODULE OF STRUCTFUSE-NET

Wavelet	OA(%)	mIou(%)
db1	**82.75**	**54.5**
db4	81.4	53.6
sym4	81.0	53.4
coif1	80.3	52.9

The superior performance of db1 can be attributed to its compact support and first-order vanishing moment, which enable efficient edge extraction and separation of high-frequency structural components from low-frequency background features [53], [54]. Unlike low-level enhancement, where wavelets primarily refine textures, high-level wavelet decomposition captures semantic-level boundaries and inter-class transitions, thereby reinforcing class-specific structural coherence. In addition, the inherent multiscale property of the db1 wavelet facilitates the joint representation of local structural details and global semantic consistency, resulting in more stable and geometry-aware fusion across modalities.

Overall, the experimental and visualization results confirm that the db1 wavelet best preserves semantic boundary integrity and interclass geometry, enhancing high-level structural fidelity and contributing to geometry-consistent optical–SAR segmentation within StructFuse-Net.

3) Effectiveness With Loss Functions: To evaluate the influence of different loss components, an ablation study was performed by incrementally adding each term in the multiobjective adaptive loss. The quantitative results on the WHU-OPT-SAR dataset are reported in Table VII. When trained solely with the L_{CE}, StructFuse-Net achieves an OA of 78.9% and a mean IoU of 47.3%, indicating that pixel-wise supervision alone is insufficient for reliable multimodal segmentation. Introducing the L_{Dice} leads to a clear performance gain of 1.9% in OA and 4.2% in mIoU, suggesting that region-level overlap optimization effectively alleviates the class-imbalance problem and stabilizes the training process.

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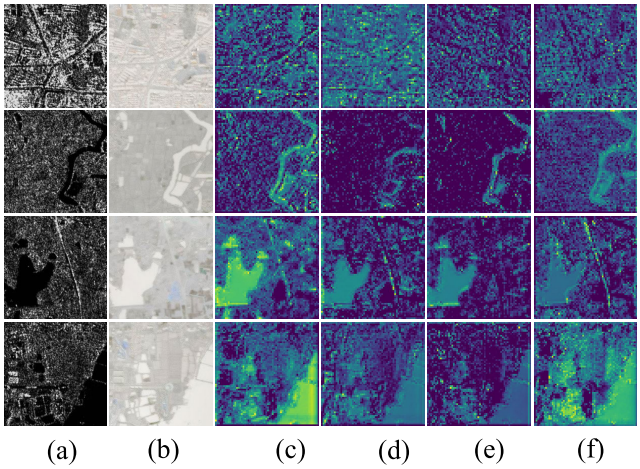


Fig. 12. Heatmap visualizations of low-level feature enhancement using different wavelet bases in the SDDF module of StructFuse-Net. (a) SAR, (b) optical, (c) db1, (d) db4, (e) sym4, and (f) coif1. The db1 wavelet preserves local geometric structures and edge continuity most effectively, whereas db4 and sym4 exhibit slightly blurred boundaries and coif1 emphasizes global texture but weakens linear continuity.

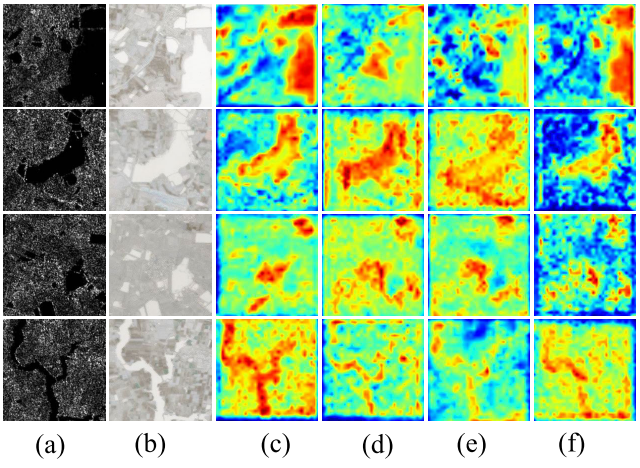


Fig. 13. Heatmap visualizations of high-level semantic feature enhancement using different wavelet bases in the SRAF module. (a) SAR, (b) optical, (c) db1, (d) db4, (e) sym4, and (f) coif1. The db1 wavelet maintains continuous semantic activations along linear targets and structured boundaries, while db4 and sym4 yield slightly weaker responses, and coif1 shows limited contribution to global semantic consistency.

TABLE VI

COMPARISON OF DIFFERENT WAVELET BASES USED FOR HIGH-LEVEL ENHANCEMENT IN THE SRAF MODULE OF STRUCTFUSE-NET

Wavelet	OA(%)	mIou(%)
db1	**82.75**	**54.5**
db4	81.9	53.7
sym4	81.65	53.5
coif1	81.4	53.2

TABLE VII

ABLATION ANALYSIS OF DIFFERENT LOSS FUNCTION COMBINATIONS ON THE WHU-OPT-SAR DATASET

L_{CE}	L_{Dice}	L_{Focal}	$L_{Boundary}$	OA(%)	mIoU(%)
✓				78.9	47.3
✓	✓			80.8	51.5
✓	✓	✓		81.8	53.2
✓	✓	✓	✓	**82.7**	**54.5**

After incorporating the L_{Focal}, the mIoU further increases to 53.2%, which reflects better handling of hard or ambiguous pixels with low confidence. The addition of the $L_{Boundary}$ brings the most notable improvement, boosting OA to 82.7% and mIoU to 54.5%. This indicates that contour-aware supervision contributes to more precise edge localization and enhances the geometric consistency of the final segmentation maps. Overall, the progressive improvement across all configurations confirms that combining complementary loss functions enables a balanced optimization between semantic accuracy and geometric fidelity within the proposed adaptive weighting framework.

In summary, the ablation results demonstrate that the proposed multiobjective adaptive loss effectively balances pixel-level, region-level, and boundary-level supervision, enabling StructFuse-Net to achieve stable optimization and geometry-consistent segmentation across multimodal data.

E. Evaluation of Model Size and Computational Cost

We further analyzed the tradeoff between segmentation accuracy and computational efficiency by comparing StructFuse-Net with a series of representative encoder designs on the WHU-OPT-SAR dataset. The evaluated models include CNN-based architectures (DeepLabV3 +, SA-Gate, MCANet, and CMFNet), transformer or hybrid architectures (CMGFNet and FTransUNet), and state-space models (RS3Mamba). As summarized in Table VIII, StructFuse-Net attains the highest

TABLE VIII

COMPARISON OF PARAMETER SIZE, COMPUTATIONAL COST, AND OA OF DIFFERENT ENCODER ARCHITECTURES ON THE WHU-OPT-SAR DATASET

Method	Parameters(M)	FLOPs(G)	OA(%)
Deeplabv3+	102	77.1	77.4
SA-Gate	110.85	**41.28**	76.1
MCANet	72.03	88.69	78.1
CMFNet	123.63	78.25	77.5
CMGFNet	64.2	19.51	77.9
RS3Mamba	**43.32**	51.65	79.1
FTransUNet	160.88	45.21	80.5
StructFuse-Net	68.8	75.1	**82.7**

OA (82.7%) while maintaining a moderate complexity of 68.8 M parameters and 75.1 GFLOPs.

Compared with heavy transformer-based encoders such as FTransUNet with 160.9 M parameters, our model achieves a relative reduction of more than 55% in parameters with a 2.2% accuracy improvement. Although CNN-based designs (SA-Gate and MCANet) exhibit lower computational costs, their performance remains limited by the lack of long-range dependency modeling. In contrast, StructFuse-Net benefits from its structure-aware multimodal fusion mechanism, which enhances global context perception without substantially increasing computational burden.

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V. Conclusion

In this work, we proposed StructFuse-Net, a structure-aware multimodal fusion framework for geometry-consistent optical–SAR RSI segmentation. Unlike conventional fusion strategies that primarily emphasize semantic aggregation, StructFuse-Net explicitly models structural consistency and geometric complementarity across heterogeneous modalities through a hierarchical design. Specifically, the SDDF module establishes robust low-level geometric alignment via bidirectional cross-modal attention and directional frequency decomposition. The SRAF module refines and aligns high-level semantic representations through channel–spatial recalibration and frequency-aware enhancement, while the SACLF module bridges cross-level representations using adaptive gating and orthogonal directional encoding to preserve multiscale structural continuity.

Extensive experiments on the WHU-OPT-SAR and YESeg-OPT-SAR datasets demonstrate that StructFuse-Net consistently outperforms SOTA methods in terms of OA and mean IoU, particularly for elongated and structure-sensitive objects such as roads and rivers. Furthermore, experiments on the cloud-disturbed Pohang dataset verify the robustness and adaptability of StructFuse-Net under severe optical degradation. Despite strong cloud-induced interference, StructFuse-Net exhibits smaller performance degradation and maintains coherent geometry and boundary continuity, highlighting the effectiveness of structure-aware and geometry-consistent fusion in challenging sensing conditions.

Despite its strong performance, StructFuse-Net incurs a relatively high computational cost (approximately 75 GFLOPs), mainly due to hierarchical structural reasoning and high-resolution cross-modal alignment. Future work will explore more efficient and lightweight fusion designs, including dynamic gating and model compression techniques, to reduce computational complexity while maintaining segmentation accuracy and geometric fidelity.

In the future, we plan to incorporate temporal alignment mechanisms and sensor-level physical priors to further enhance robustness and physical interpretability. Integrating SAR acquisition metadata and handling nonsynchronous multimodal observations will enable broader generalization to practical geoscience applications, such as multimodal change detection, environmental monitoring, and disaster assessment.

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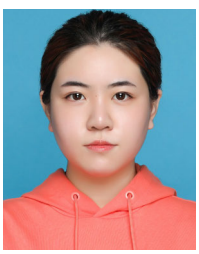
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